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Review article

Robust control under parametric uncertainty: An overview and recent results

S.P. Bhattacharyya

Department of Electrical and Computer Engineering, Texas A & M University, College Station, TX 77843, USA

A R T I C L E I N F O

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A B S T R A C T

Modern Robust Control has had two distinct lines of development: (a) Robustness through quadratic optimization and (b) Robustness under parametric uncertainty. The first approach consists of Kalman's Linear Quadratic Regulator and H_∞ optimal control. The second approach is the focus of this overview paper. It provides an account of both analysis as well as synthesis based results. This line of results was sparked by the appearance of Kharitonov's Theorem in the early1980s. This result was rapidly followed by further results on the stability of polytopes of polynomials such as the Edge Theorem and the Generalized Kharitonov Theorem, stability of systems under norm bounded perturbations and the computation of parametric stability margins. Many of these analysis results established extremal testing sets where stability or performance would breakdown. Starting in 1997, when it was established that high order controllers were fragile, attention turned to the synthesis and design of the parameters of low order controllers such as three term controllers and more particularly Proportional-Integral-Derivative (PID) controllers. An extensive theory of design of such systems has developed in the last twenty years. We provide a summary without proofs, of many of these results.

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E-mail addresses: spb@tamu.edu, spb_tx@yahoo.com

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1. Introduction

Robustness of a system, the subject of this article, is its ability to remain functional despite large changes. In control engineering, robustness has played a central and pivotal role, since its beginning in the 1860s. Thus Black's feedback amplifier (Kline, 1993), the Nyquist criterion (Nyquist, 1932), and gain and phase margins Bode (1945) were concepts dealing directly with robustness in the classical period.

Starting in 1960, the focus of control engineers shifted to optimization. However, the adequacy of an optimal design was ultimately judged by its robustness. Kalman's Linear Quadratic Optimal Regulator (Kalman, 1959) was found to be deficient when measured by its ability to deliver stability margins under output feedback (Doyle & Stein, 1979). The remedy proposed was high order H_∞ control (Doyle, Glover, Khargonekar, & Francis, 1989). In 1997, (Keel & Bhattacharyya, 1997) it was shown that even these controllers, and indeed all high order controllers, were fragile. This led of a renewed interest in direct studies on robustness resulting in a body of knowledge known as the parametric theory (Ackermann, 2012; Barmish & Jury, 1994; Bhattacharyya, Chapellat, & Keel, 1995; Bhattacharyya, 1987). This theory has two components: analysis and synthesis. The present paper gives an overview account of the analysis results, Kharitonov's theorem and its generalization (Chapellat & Bhattacharyya, 1989; Kharitonov, 1978), the Edge theorem (Bartlett, Hollot, & Lin, 1988), and related results as well as recent results on the parametric theory of synthesis and design (Bhattacharyya et al., 1995) of Proportional-Integral-Derivative (PID) controllers, Datta, Ho, and Bhattacharyya (2013), Silva, Datta, and Bhattacharyya (2007), Diaz-Rodriguez, Oliveira, and Bhattacharyya (2015), Diaz-Rodriguez and Bhattacharyya (2015).

1.1. Quadratic optimization and robustness

In Kalman et al. (1960) introduced the state-variable approach and quadratic optimal control in the time-domain as new design approaches. This phase in the theory of automatic control systems arose out of the important new technological problems that were encountered at that time: the launching, maneuvering, guidance and tracking of space vehicles. A lot of effort was expended and rapid developments in both theory and practice took place. Optimal control theory was developed under the influence of many great researchers such as Pontryagin, Bellman, Kalman and Bucy. In the 1960s, Kalman introduced a number of key state-variable concepts. Among these were controllability, observability, optimal linear-quadratic regulator (LQR), state-feedback and optimal state estimation (Kalman filtering).

The optimal state feedback control produced by the LQR problem was guaranteed to be stabilizing for any quadratic performance index subject to mild conditions.

In a 1964 paper by Kalman (1964) which demonstrated that for SISO (single input-single output) systems the optimal LQR state-feedback control laws had some very strong guaranteed robustness properties, namely an infinite upper gain margin and a 60 ° phase margin, which in addition were independent of the particular quadratic index chosen. This is illustrated in Fig. 1 where the state feedback system designed via LQR optimal control has the above guaranteed stability margins at the loop breaking point "*m*".

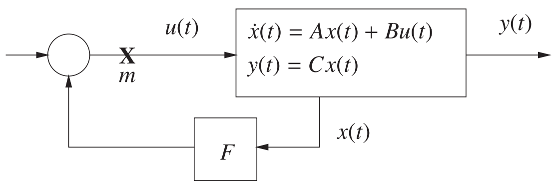

Fig. 1. State feedback configuration.

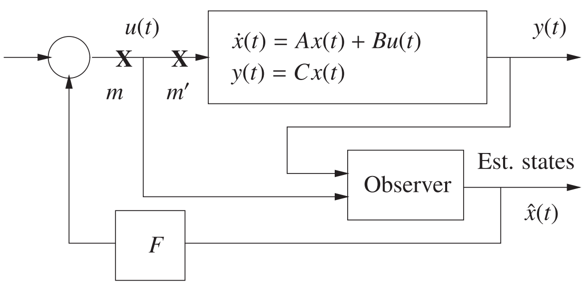

Fig. 2. Observed state feedback configuration.

For some time, control scientists were generally led to believe that the extraordinary robustness properties of the LQR state feedback design were preserved when the control was implemented as an output feedback system through an observer. We depict this in Fig. 2 where the stability margin at the point m continues to equal that obtained in the state feedback system. However it was shown by Doyle and Stein (1979) that the margin at the point m', which is much more meaningful, could be drastically less. This observation ushered in a period of renewed interest in robustness of closed loop designs.

1.2. Robustness under parametric uncertainty

Here we first describe a group of results which may be considered to be the central results of analysis in the field parametric theory of robust control. They are characterized by the important feature that they facilitate robust stability calculations by identifying *apriori* a small subset of parameters where stability or performance will be lost. Proofs of most of the results described here can be found in (Bhattacharyya et al., 1995; Bhattacharyya, Datta, & Keel, 2009). We begin with the most spectacular of these results, namely Kharitonov's Theorem Kharitonov (1978), which gives a surprisingly simple necessary and sufficient condition for the robust stability of an interval family of polynomials.

In 1997 it was shown that high order controllers were acutely sensitive to controller parameter perturbations. This led to a resurgence of interest in 3 term controllers, in particular PID controllers. An extensive theory of synthesis and design of PID controllers has been developed over the last 20 years (Bhattacharyya et al., 1995; Datta et al., 2013; Diaz-Rodriguez & Bhattacharyya, 2015; Diaz-Rodriguez et al., 2015; Silva et al., 2007). We give an account of these elegant and useful results in the last part of the paper.

2. Kharitonov's theorem

Consider the set $\mathcal{I}(s)$ of polynomials of degree n with real coefficients of the form

$$\delta(s) = \delta_o + \delta_1 s + \delta_2 s^2 + \delta_3 s^3 + \delta_4 s^4 + \cdots + \delta_n s^n$$

where the coefficients lie within given ranges,

$$\delta_0 \in [x_0, y_0], \;\; \delta_1 \in [x_1, y_1], \;\; \ldots, \;\; \delta_n \in [x_n, y_n].$$

Write

$$\underline{\delta} := [\delta_0, \;\delta_1, \;\ldots, \;\delta_n]$$

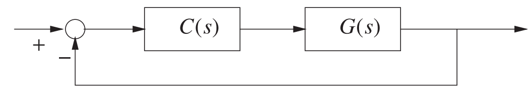

Fig. 3. Feedback system with controller (Example 2.2).

and identify a polynomial $\delta(s)$ with its coefficient vector $\underline{\delta}$. Introduce the box of coefficients

$$\Delta := \left\{ \underline{\delta} : \underline{\delta} \in \mathbb{R}^{n+1}, \;\; x_i \le \delta_i \le y_i, \;\; i = 0, 1, \ldots, n \right\}. \tag{1}$$

We assume that the degree remains invariant over the family, so that $0 \notin [x_n, \, y_n]$. Such a set of polynomials is called a real *interval* family and we loosely refer to $\mathcal{I}(s)$ or Δ as an interval polynomial.

Theorem 2.1 (Kharitonov's Theorem)**.** *Kharitonov (1978) Every polynomial in the family $\mathcal{I}(s)$ is Hurwitz if and only if the following four* extreme *polynomials are Hurwitz:*

$$\begin{aligned} K^1(s) &= x_o + x_1 s + y_2 s^2 + y_3 s^3 + x_4 s^4 + \cdots, \\ K^2(s) &= x_o + y_1 s + y_2 s^2 + x_3 s^3 + x_4 s^4 + \cdots, \\ K^3(s) &= y_o + x_1 s + x_2 s^2 + y_3 s^3 + y_4 s^4 + \cdots, \\ K^4(s) &= y_o + y_1 s + x_2 s^2 + x_3 s^3 + y_4 s^4 + \cdots. \end{aligned} \tag{2}$$

The surprising element in Kharitonov's Theorem is due to the fact that the Hurwitz region in coefficient space is *not* convex, and that the number four of test polynomials is *independent* of n. The following example illustrates its applicability to a control system.

Example 2.2. Consider the control system shown in Fig. 3. The plant is described by the rational transfer function $G(s)$ with numerator and denominator coefficients varying independently in prescribed intervals. We refer to such a family of transfer functions $\mathbf{G}(s)$ as an *interval plant*. In the present example we take

$$\mathbf{G}(s) := \left\{ G(s) = \frac{n_2 s^2 + n_1 s + n_0}{s^3 + d_2 s^2 + d_1 s + d_0} \right\} \tag{3}$$

where

$$\begin{array}{lll} n_0 \in [1, 2.5], & n_1 \in [1, 6], & n_2 \in [1, 7], \\ d_2 \in [-1, 1], & d_1 \in [-0.5, 1.5], & d_0 \in [1, 1.5]. \end{array}$$

The controller is a constant gain, $C(s) = k$ that is to be adjusted, if possible, to robustly stabilize the closed loop system. More precisely we are interested in determining the range of values of the gain $k \in [-\infty, +\infty]$ for which the closed loop system is robustly stable, that is, stable for all $G(s) \in \mathbf{G}(s)$.

The characteristic polynomial of the closed loop system is:

$$\delta(k, s) = s^3 + \underbrace{(d_2 + kn_2)}_{\delta_2(k)} s^2 + \underbrace{(d_1 + kn_1)}_{\delta_1(k)} s + \underbrace{(d_0 + kn_0)}_{\delta_0(k)}.$$

Since the parameters $d_i, \; n_j, \; i = 0, 1, 2, \;\; j = 0, 1, 2$ vary independently it follows that for each fixed k, $\delta(k, \, s)$ is an interval polynomial. Using the bounds given to describe the family $\mathbf{G}(s)$ we get the following coefficient bounds for positive k:

$$\begin{aligned} &\delta_2(k) \in [-1 + k, 1 + 7k], \\ &\delta_1(k) \in [-0.5 + k, 1.5 + 6k], \\ &\delta_0(k) \in [1 + k, 1.5 + 2.5k]. \end{aligned}$$

By applying Kharitonov's Theorem we get the necessary and sufficient conditions for robust stability to be:

$$\begin{array}{lll} -1 + k > 0, & -0.5 + k > 0, & 1 + k > 0, \\ (-0.5 + k)(-1 + k) > 1.5 + 2.5k. & & \end{array}$$

From this it follows that the closed loop system is robustly stable if and only if

$$k \in (2 + \sqrt{5}, +\infty).$$

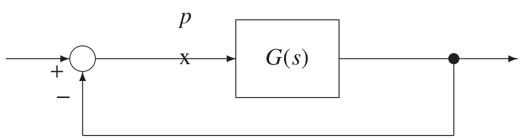

Fig. 4. A feedback system.

3. Extremal properties of edges and vertices

In this section we state some useful extremal properties of the Kharitonov polynomials. Suppose that we have proved the stability of the family of polynomials

$$\delta(s)=\delta_0+\delta_1 s+\delta_2 s^2+\cdots+\delta_n s^n,$$

with coefficients in the box

$$\mathbf{\Delta}=[x_0,y_0]\times[x_1,y_1]\times\cdots\times[x_n,y_n].$$

Each polynomial in the family is stable. A natural question that arises now is the following: What point in $\mathbf{\Delta}$ is closest to instability? The stability margin of this point is in a sense the worst case stability margin of the interval system. It turns out that a precise answer to this question can be given in terms of the parametric stability margin as well as in terms of the gain margins of an associated interval system Bhattacharyya et al. (1995). We first deal with the extremal parametric stability margin problem.

3.1. Extremal parametric stability margin property

We consider a stable interval polynomial family. It is therefore possible to associate with each polynomial of the family the largest stability ball centered around it.

Write

$$\underline{\delta}=[\delta_0,\delta_1,\ldots,\delta_n],$$

and regard $\underline{\delta}$ as a point in $\mathbb{R}^{n+1}$. Let $\|\ \underline{\delta}\ \|_p$ denote the p norm in $\mathbb{R}^{n+1}$ and let this be associated with $\delta(s)$. The set of polynomials which are unstable of degree n or of degree less than n is denoted by $\mathcal{U}$. Then the radius of the stability ball centered at $\boldsymbol{\delta}$ is

$$\rho(\delta)=\inf_{u\in\mathcal{U}}\|\underline{\delta}-\mathbf{u}\|_p.$$

We thus define a mapping from $\mathbf{\Delta}$ to the set of all positive real numbers:

$$\mathbf{\Delta}\xrightarrow{\rho}\mathcal{R}^+\backslash\{0\}$$
$$\delta(s)\longrightarrow\rho(\delta).$$

A natural question to ask is the following: Is there a point in $\mathbf{\Delta}$ which is the nearest to instability? Or stated in terms of functions: Has the function ρ a minimum and is there a precise point in $\mathbf{\Delta}$ where it is reached? The answer to that question is given in the following theorem. In the discussion to follow we drop the subscript p from the norm since the result holds for any norm chosen.

Theorem 3.1 (Extremal property of the Kharitonov polynomials)**.** Bhattacharyya et al. (1995) *The function*

$$\mathbf{\Delta}\xrightarrow{\rho}\mathcal{R}^+\backslash\{0\}$$
$$\delta(s)\longrightarrow\rho(\delta)$$

has a minimum which is reached at one of the four Kharitonov polynomials associated with $\mathbf{\Delta}$.

3.2. Extremal gain margin for interval systems

Let us now consider the standard unity feedback control system shown below in Fig. 4. We assume that the system represented by $G(s)$ contains parameter uncertainty. In particular let us assume that $G(s)$ is a proper transfer function which is a ratio of polynomials $n(s)$ and $d(s)$ the coefficients of which vary in independent intervals. Thus, the polynomials $n(s)$ and $d(s)$ vary in respective independent interval polynomial families $\mathcal{N}(s)$ and $\mathcal{D}(s)$ respectively. We refer to such a family of systems as an *interval system*. Let

$$\mathbf{G}(s):=\left\{G(s)=\frac{n(s)}{d(s)}:n(s)\in\mathcal{N}(s),\quad d(s)\in\mathcal{D}(s)\right\}.$$

represent the interval family of systems in which the open-loop transfer function lies.

We assume that the closed-loop system containing the interval family $\mathbf{G}(s)$ is robustly stable. In other words we assume that the characteristic polynomial of the closed-loop system given by

$$\Pi(s)=d(s)+n(s)$$

is of invariant degree n and is Hurwitz for all $(n(s),d(s))\in(\mathcal{N}(s)\times\mathcal{D}(s))$. Let

$$\mathbf{\Pi}(s)=\{\Pi(s)=n(s)+d(s):n(s)\in\mathcal{N}(s),\quad d(s)\in\mathcal{D}(s)\}.$$

Then robust stability means that every polynomial in $\mathbf{\Pi}(s)$ is Hurwitz and of degree n. This can in fact be verified constructively. Let $K_N^i(s), i=1,2,3,4$ and $K_D^j(s), j=1,2,3,4$ denote the Kharitonov polynomials associated with $\mathcal{N}(s)$ and $\mathcal{D}(s)$ respectively. Now introduce the positive set of *Kharitonov systems*$\mathbf{G}_{\mathrm{K}}^+(s)$ associated with the interval family $\mathbf{G}(s)$ as follows:

$$\mathbf{G}_{\mathrm{K}}^+(s):=\left\{\frac{K_N^i(s)}{K_D^i(s)}:i=1,2,3,4\right\}.$$

Theorem 3.2. Bhattacharyya et al. (1995) *The closed-loop system of Fig. 4 containing the interval plant* $\mathbf{G}(s)$ *is robustly stable if and only if each of the positive Kharitonov systems in* $\mathbf{G}_K^+(s)$ *is stable.*

In classical control *gain margin* is commonly regarded as a useful measure of robust stability. Suppose the closed system is stable. The gain margin at the loop breaking point p is defined to be the largest value k^* of $k\geq 1$ for which closed-loop stability is preserved with $G(s)$ replaced by $kG(s)$ for all $k\in[1,k^*)$. Suppose now that we have verified the robust stability of the interval family of systems $\mathbf{G}(s)$. The next question that is of interest is: What is the gain margin of the system at the loop breaking point p? To be more precise we need to ask: What is the *worst case* gain margin of the system at the point p as $G(s)$ ranges over $\mathbf{G}(s)$? An exact answer to this question can be given as follows.

Theorem 3.3. Bhattacharyya et al. (1995) *The worst case gain margin of the system at the point p over the family* $\mathbf{G}(s)$ *is the minimum gain margin corresponding to the positive Kharitonov systems* $\mathbf{G}_{\mathrm{K}}^+(s)$.

4. Robust state feedback stabilization

In this section we give an application of Kharitonov's Theorem to robust stabilization by state feedback. We consider the following problem: Suppose that you are given a set of n nominal parameters

$$\left\{a_0^0,a_1^0,\ldots,a_{n-1}^0\right\},$$

together with a set of prescribed uncertainty ranges: Δa_0, Δa_1, $\ldots$, Δa_{n-1}, and that you consider the family $\mathcal{I}_{\underline{0}}(s)$ of monic polynomials,

$$\delta(s)=\delta_0+\delta_1 s+\delta_2 s^2+\cdots+\delta_{n-1}s^{n-1}+s^n,$$

where

$$\delta_0\in\left[a_0^0-\frac{\Delta a_0}{2},a_0^0+\frac{\Delta a_0}{2}\right],\ldots,$$

$$\delta_{n-1} \in \left[a_{n-1}^0 - \frac{\Delta a_{n-1}}{2}, a_{n-1}^0 + \frac{\Delta a_{n-1}}{2}\right].$$

To avoid trivial cases assume that the family $\mathcal{I}_{\underline{0}}(s)$ contains at least one unstable polynomial.

Suppose now that you can use a vector of n free parameters

$$\underline{k} = (k_0, k_1, \ldots, k_{n-1}),$$

to transform the family $\mathcal{I}_{\underline{0}}(s)$ into the family $\mathcal{I}_{\underline{k}}(s)$ described by:

$$\delta(s) = (\delta_0 + k_0) + (\delta_1 + k_1)s + (\delta_2 + k_2)s^2 + \cdots + (\delta_{n-1} + k_{n-1})s^{n-1} + s^n.$$

The problem of interest, then, is the following: Given Δa_0, Δa_1, …, Δa_{n-1} the perturbation ranges fixed a priori, find, if possible, a vector $\underline{k}$ so that the new family of polynomials $\mathcal{I}_{\underline{k}}(s)$ is entirely Hurwitz stable. This problem arises, for example, when one applies a state-feedback control to a single input system where the matrices A, b are in controllable companion form and the coefficients of the characteristic polynomial of A are subject to bounded perturbations. The answer to this problem is always affirmative and is precisely given in Theorem 4.1. We can now enunciate the following theorem to answer the question raised at the beginning of this section.

Theorem 4.1. *Bhattacharyya et al. (1995) For any set of nominal parameters $\{a_0, a_1, \ldots, a_{n-1}\}$, and for any set of positive numbers Δa_0, Δa_1, …, Δa_{n-1}, it is possible to find a vector $\underline{k}$ such that the entire family $\mathcal{I}_{\underline{k}}(s)$ is stable.*

Example 4.2. Suppose that our nominal polynomial is

$$s^6 - s^5 + 2s^4 - 3s^3 + 2s^2 + s + 1,$$

that is

$$(a_0^0, a_1^0, a_2^0, a_3^0, a_4^0, a_5^0) = (1, 1, 2, -3, 2, -1).$$

And suppose that we want to handle the following set of uncertainty ranges:

$$\Delta a_0 = 3, \quad \Delta a_1 = 5, \quad \Delta a_2 = 2, \quad \Delta a_3 = 1, \quad \Delta a_4 = 7, \quad \Delta a_5 = 5.$$

Step 1: Consider the following stable polynomial of degree 5

$$R(s) = (s+1)^5 = 1 + 5s + 10s^2 + 10s^3 + 5s^4 + s^5.$$

The calculation of $\rho(R(\ \cdot\ ))$ gives: $\rho(R(\cdot)) = 1$. On the other hand we have

$$\sqrt{\frac{{\Delta a_0}^2}{4} + \frac{{\Delta a_1}^2}{4} + \cdots + \frac{{\Delta a_{n-1}}^2}{4}} = 5.31.$$

Taking therefore $\lambda = 6$, we have that

$$P(s) = 6 + 30s + 60s^2 + 60s^3 + 30s^4 + 6s^5,$$

has a radius $\rho(P(\cdot)) = 6$ that is greater than 5.31. The four polynomials $P^j(s)$ are given by

$$P^1(s) = 4.5 + 27.5s + 61s^2 + 60.5s^3 + 26.5s^4 + 3.5s^5,$$
$$P^2(s) = 4.5 + 32.5s + 61s^2 + 59.5s^3 + 26.5s^4 + 8.5s^5,$$
$$P^3(s) = 7.5 + 27.5s + 59s^2 + 60.5s^3 + 33.5s^4 + 3.5s^5,$$
$$P^4(s) = 7.5 + 32.5s + 59s^2 + 59.5s^3 + 33.5s^4 + 8.5s^5.$$

Step 2: The application of Lemma 11.7 in Bhattacharyya et al. (2009) gives the following values

$$\alpha_1 \simeq 1.360, \quad \alpha_2 \simeq 2.667, \quad \alpha_3 \simeq 1.784, \quad \alpha_4 \simeq 3.821,$$

and therefore we can chose $\alpha = 1$, so that the four polynomials

$$K^1(s) = 4.5 + 27.5s + 61s^2 + 60.5s^3 + 26.5s^4 + 3.5s^5 + s^6,$$
$$K^2(s) = 4.5 + 32.5s + 61s^2 + 59.5s^3 + 26.5s^4 + 8.5s^5 + s^6,$$
$$K^3(s) = 7.5 + 27.5s + 59s^2 + 60.5s^3 + 33.5s^4 + 3.5s^5 + s^6,$$
$$K^4(s) = 7.5 + 32.5s + 59s^2 + 59.5s^3 + 33.5s^4 + 8.5s^5 + s^6,$$

are stable.

Step 3: We just have to take

$$k_0 = p_0 - a_0 = 5, \quad k_1 = p_1 - a_1 = 29, \quad k_2 = p_2 - a_2 = 58,$$
$$k_3 = p_3 - a_3 = 63, \quad k_4 = p_4 - a_4 = 28, \quad k_5 = p_5 - a_5 = 7.$$

to robustly stabilize the system.

5. The edge theorem

The interval family dealt with in Kharitonov's Theorem is a very special type of polytopic family. Moreover Kharitonov's Theorem does not indicate where the roots of the polynomial family lie. The Edge Theorem deals with a general convex polytopic family of polynomials and gives a complete, exact and constructive characterization of the root set of the family. Such a characterization is obviously of value in the robustness and performance analysis of control systems. We describe this remarkable theorem below.

Consider a family of nth degree real polynomials whose typical element is given by

$$\delta(s) = \delta_0 + \delta_1 s + \cdots + \delta_{n-1}s^{n-1} + \delta_n s^n. \tag{4}$$

We will identify the polynomial in (4) with the vector

$$\underline{\delta} := [\delta_n, \delta_{n-1}, \cdots, \delta_1, \delta_0]^T. \tag{5}$$

Let $\Omega \subset \mathbb{R}^{n+1}$ be an m-dimensional *polytope*, that is, the convex hull of a finite number of points. We make the assumption that all polynomials in Ω have the same degree and the sign of δ_n is constant over Ω, say, positive.

A *supporting hyperplane* H is an affine set of dimension n such that $\Omega \cap H \neq \emptyset$, and such that every point of Ω lies on just one side of H. The *exposed sets* of Ω are those (convex) sets $\Omega \cap H$ where H is a supporting hyperplane. The one dimensional exposed sets are called *exposed edges*.

For any $W \subset \Omega$, $R(W)$ is said to be the root space of W if,

$$R(W) = \{s : \delta(s) = 0, \quad \text{for some } \underline{\delta} \in W\}. \tag{6}$$

Finally, let the boundary of an arbitrary set S of the complex plane be designated by ∂S. We can now enunciate the Edge Theorem, developed by Bartlett et al. (1988).

Theorem 5.1 (Edge theorem)**.** *Bartlett et al. (1988) Let $\Omega \subset \mathbb{R}^{n+1}$ be a polytope of polynomials as above. Then the boundary of $R(\Omega)$ is contained in the root space of the exposed edges of Ω.*

We illustrate the importance of this result in control systems with an example.

Example 5.2. Consider the unity feedback discrete time control system with forward transfer function:

$$G(z) = \frac{\delta_1 z + \delta_0}{z^2(z + \delta_2)}.$$

The characteristic polynomial is

$$\delta(z) = z^3 + \delta_2 z^2 + \delta_1 z + \delta_0.$$

Suppose that the coefficients vary in the intervals

$$\delta_2 \in [0.042, 0.158], \quad \delta_1 \in [-0.058, 0.058],$$
$$\delta_0 \in [-0.06, 0.056]$$

The boundary of the root space of the family can be generated by drawing the root loci along the 12 exposed edges of the box in coefficient space. The root space is inside the unit disc as shown in Fig. 5. Hence the entire family is Schur stable.

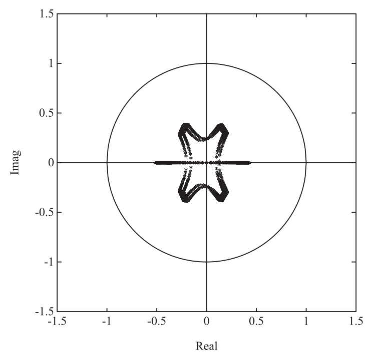

Fig. 5. Root space of $\delta(z)$ (Example 5.2).

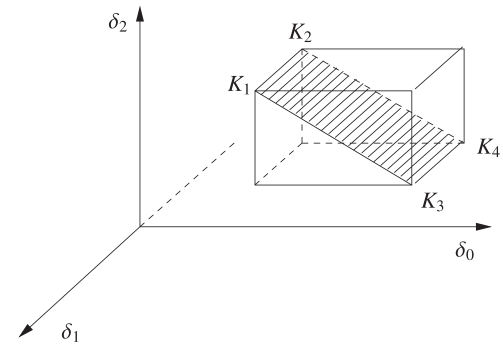

Fig. 6. The four Kharitonov segments.

6. The generalized kharitonov theorem

Kharitonov's Theorem applies to polynomial families where the coefficients vary *independently*. In a typical control system problem, the closed loop characteristic polynomial coefficients vary *interdependently*. For example the closed loop characteristic polynomial coefficients may vary only through the perturbation of the plant parameters while the controller parameters remain fixed. The Generalized Kharitonov Theorem described below, deals with this situation and develops results that retain the important feature of Kharitonov's Theorem, namely that the number of items to be checked for stability does not increase with number of parameters. In this respect it is important to remember that the Edge Theorem has the drawback that it requires checking of all the exposed edges and this number increases exponentially with the number of parameters.

Consider real characteristic polynomials of the form

$$\delta(s) = F_1(s)P_1(s) + \cdots + F_m(s)P_m(s). \tag{7}$$

Write

$$\underline{F}(s) := (F_1(s), F_2(s), \cdots\cdots F_m(s)) \tag{8}$$

$$\underline{P}(s) := (P_1(s), P_2(s), \ldots, P_m(s)) \tag{9}$$

and introduce the notation

$$\langle \underline{F}(s), \underline{P}(s)\rangle := F_1(s)P_1(s) + \cdots + F_m(s)P_m(s). \tag{10}$$

We will say that $\underline{F}(s)$ *stabilizes* $\underline{P}(s)$ if $\delta(s) =< \underline{F}(s), \underline{P}(s) >$ is *Hurwitz stable*. In this subsection, stable will mean Hurwitz stable.

The polynomials $F_i(s)$ are fixed real polynomials whereas $P_i(s)$ are real polynomials with coefficients varying independently in prescribed intervals.

Let $d^o(P_i)$ be the degree of $P_i(s)$

$$P_i(s) := p_{i,0} + p_{i,1}s + \cdots + p_{i,d^o(P_i)}s^{d^o(P_i)}. \tag{11}$$

and

$$\mathbf{p}_i := [p_{i,0}, p_{i,1}, \cdots, p_{i,d^o(P_i)}]. \tag{12}$$

Let $\underline{n} = [1, 2, \ldots, n]$. Each $P_i(s)$ belongs to an interval family $\mathbf{P}_i(s)$ specified by the intervals

$$p_{i,j} \in [\alpha_{i,j}, \beta_{i,j}] \quad i \in \underline{m} \quad j = 0, \cdots, d^o(P_i). \tag{13}$$

The corresponding parameter box is

$$\Pi_i := \{\mathbf{p}_i : \alpha_{i,j} \le p_{i,j} \le \beta_{i,j}, j = 0, 1, \ldots, d^o(P_i)\}. \tag{14}$$

Let us denote the corresponding family of polynomial m-tuples

$$\mathbf{P}(s) := \mathbf{P}_1(s) \times \mathbf{P}_2(s) \times \cdots \times \mathbf{P}_m(s). \tag{15}$$

Let

$$\mathbf{p} := [\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_m] \tag{16}$$

denote the global parameter vector and let

$$\Pi := \Pi_1 \times \Pi_2 \times \cdots \times \Pi_m \tag{17}$$

denote the global parameter uncertainty set. Now let us consider the polynomial (7) and rewrite it as $\delta(s, \mathbf{p})$ or $\delta(s, \underline{P}(s))$ to emphasize its dependence on the parameter vector $\mathbf{p}$ or the m-tuple $\underline{P}(s)$. We are interested in determining the Hurwitz stability of the set of polynomials

$$\begin{aligned}\Delta(s) &:= \{\delta(s, \mathbf{p}) : \mathbf{p} \in \Pi\} \\ &= \{\langle \underline{F}(s), \underline{P}(s)\rangle : \underline{P}(s) \in \mathbf{P}(s)\}.\end{aligned} \tag{18}$$

We call this a *linear interval polynomial* and adopt the notational convention

$$\Delta(s) = F_1(s)\mathbf{P}_1(s) + \cdots + F_m(s)\mathbf{P}_m(s). \tag{19}$$

We shall make the standing assumptions that the elements of $\mathbf{p}$ perturb independently of each other and that every polynomial in $\Delta(s)$ is of the same degree. Henceforth we will say that $\Delta(s)$ *is stable if every polynomial in* $\Delta(s)$ *is Hurwitz stable*. An equivalent statement is that $\underline{F}(s)$ *stabilizes every* $\underline{P}(s) \in \mathbf{P}(s)$.

The solution described below constructs an extremal set of line segments $\Delta_E(s) \subset \Delta(s)$ with the property that the stability of $\Delta_E(s)$ implies stability of $\Delta(s)$. This solution is constructive because the stability of $\Delta_E(s)$ can be checked, for instance by a set of root locus problems. The solution is efficient since the number of elements of $\Delta_E(s)$ will be independent of the dimension of the parameter space Π. The extremal subset $\Delta_E(s)$ will be generated by first constructing an extremal subset $\mathbf{P}_E(s)$ of the m-tuple family of plants $\mathbf{P}(s)$. The extremal subset $\mathbf{P}_E(s)$ is constructed from the Kharitonov polynomials of $\mathbf{P}_i(s)$. We describe this construction next.

Construction of the extremal subset

The Kharitonov polynomials corresponding to each $\mathbf{P}_i(s)$ are denoted as:

$$\mathcal{K}_i(s) := \{K_i^1(s), K_i^2(s), K_i^3(s), K_i^4(s)\}. \tag{20}$$

For each $\mathbf{P}_i(s)$ we introduce 4 line segments, or convex combinations, joining pairs of Kharitonov polynomials as defined below:

$$\mathcal{S}_i(s) := \left\{ \begin{array}{l} \left[K_i^1(s), K_i^2(s)\right], \left[K_i^1(s), K_i^3(s)\right], \\ \left[K_i^2(s), K_i^4(s)\right], \left[K_i^3(s), K_i^4(s)\right] \end{array} \right\}. \tag{21}$$

These four segments are called *Kharitonov segments*. They are illustrated in Fig. 6 for the case of a polynomial of degree 2. For each $l \in \{1, \ldots, m\}$ let us define

$$\begin{aligned}&\mathbf{P}_E^l(s) := \\ &\mathcal{K}_1(s) \times \cdots \times \mathcal{K}_{l-1}(s) \times \mathcal{S}_l(s) \times \mathcal{K}_{l+1}(s) \times \cdots \times \mathcal{K}_m(s).\end{aligned} \tag{22}$$

The *extremal subset* of $\mathbf{P}(s)$ is defined by

$$\mathbf{P}_E(s) := \cup_{l=1}^m \mathbf{P}_E^l(s). \tag{23}$$

The corresponding set of *generalized Kharitonov segment* polynomials are

$$\Delta_E(s) := \cup_{l=1}^{m} \Delta_E^l(s) = \{\langle \underline{F}(s), \underline{P}(s)\rangle \ : \ \underline{P}(s) \in \mathbf{P}_E(s)\}. \tag{24}$$

The set of m-tuples of Kharitonov polynomials are denoted $\mathbf{P}_K(s)$ and referred to as the *Kharitonov vertices* of $\mathbf{P}(s)$:

$$\mathbf{P}_K(s) := \mathcal{K}_1(s) \mathrm{x} \mathcal{K}_2(s) \mathrm{x} \cdots \mathrm{x} \mathcal{K}_m(s) \subset \mathbf{P}_E(s). \tag{25}$$

The corresponding set of *Kharitonov vertex* polynomials is

$$\Delta_K(s) := \{\langle \underline{F}(s), \underline{P}(s)\rangle \ : \ \underline{P}(s) \in \mathbf{P}_K(s)\}. \tag{26}$$

A typical element of $\Delta_K(s)$ is

$$F_1(s)K_1^{j_1}(s) + F_2(s)K_2^{j_2}(s) + \cdots + F_m(s)K_m^{j_m}(s). \tag{27}$$

The set $\mathbf{P}_E(s)$ is made up of one parameter families of polynomial vectors. It is easy to see that there are $m4^m$ such segments in the most general case where there are four distinct Kharitonov polynomials for each $\mathbf{P}_i(s)$. The parameter space subsets corresponding to $\mathbf{P}_E^l(s)$ and $\mathbf{P}_E(s)$ are denoted by Π_l and

$$\Pi_E := \cup_{l=1}^{m} \Pi_l, \tag{28}$$

respectively. Similarly, let Π_K denote the vertices of Π corresponding to the Kharitonov polynomials. Then, we also have

$$\Delta_E(s) := \{\delta(s, \mathbf{p}) : \mathbf{p} \in \Pi_E\} \tag{29}$$

$$\Delta_K(s) := \{\delta(s, \mathbf{p}) : \mathbf{p} \in \Pi_K\} \tag{30}$$

The set $\mathbf{P}_K(s)$ in general has 4^m distinct elements when each $\mathbf{P}_i(s)$ has four distinct Kharitonov polynomials. Thus $\Delta_K(s)$ is a discrete set of polynomials, $\Delta_E(s)$ is a set of line segments of polynomials, $\Delta(s)$ is a polytope of polynomials, and

$$\Delta_K(s) \subset \Delta_E(s) \subset \Delta(s). \tag{31}$$

With these preliminaries, we are ready to state the Generalized Kharitonov Theorem (GKT).

Generalized Kharitonov theorem (GKT): (Chapellat & Bhattacharyya, 1989) For a given m-tuple $\underline{F}(s) = (F_1(s), \cdots, F_m(s))$ of real polynomials:

(I) $\underline{F}(s)$ stabilizes the entire family $\mathbf{P}(s)$ of m-tuples if and only if $\underline{F}$ stabilizes every m-tuple segment in $\mathbf{P}_E(s)$. Equivalently, $\Delta(s)$ is stable if and only if $\Delta_E(s)$ is stable.

(II) Moreover, if the polynomials $F_i(s)$ are of the form

$$F_i(s) = s^{t_i}(a_i s + b_i)U_i(s)Q_i(s)$$

where $t_i \geq 0$ is an arbitrary integer, a_i and b_i are arbitrary real numbers, $U_i(s)$ is an anti-Hurwitz polynomial, and $Q_i(s)$ is an even or odd polynomial, then it is enough that $\underline{F}(s)$ stabilizes the finite set of m-tuples $\mathbf{P}_K(s)$, or equivalently, that the set of Kharitonov vertex polynomials $\Delta_K(s)$ are stable.

(III) Finally, stabilizing the finite set $\mathbf{P}_K(s)$ is not sufficient to stabilize $\mathbf{P}(s)$ when the polynomials $F_i(s)$ do not satisfy the conditions in II). Equivalently, stability of $\Delta_K(s)$ does not imply stability of $\Delta(s)$ when $F_i(s)$ do not satisfy the conditions in II).

We illustrate the control application of this Theorem below.

Example 6.1. Consider the plant

$$G(s) = \frac{P_1(s)}{P_2(s)} = \frac{s^3 + \alpha s^2 - 2s + \beta}{s^4 + 2s^3 - s^2 + \gamma s + 1}$$

where

$$\alpha \in [-1, -2], \quad \beta \in [0.5, 1], \quad \gamma \in [0, 1].$$

Let

$$C(s) = \frac{F_1(s)}{F_2(s)}$$

denote the compensator. To determine if $C(s)$ robustly stabilizes the set of plants given we must verify the Hurwitz stability of the family of characteristic polynomials $\Delta(s)$ defined as

$$F_1(s)(s^3 + \alpha s^2 - 2s + \beta) + F_2(s)(s^4 + 2s^3 - s^2 + \gamma s + 1)$$

with $\alpha \in [-1, -2]$, $\beta \in [0.5, 1]$, $\gamma \in [0, 1]$. To construct the generalized Kharitonov segments, we start with the Kharitonov polynomials. There are two Kharitonov polynomials associated with $P_1(s)$

$$K_1^1(s) = K_1^2(s) = 0.5 - 2s - s^2 + s^3$$

$$K_1^3(s) = K_1^4(s) = 1 - 2s - 2s^2 + s^3$$

and also two Kharitonov polynomials associated with $P_2(s)$

$$K_2^1(s) = K_2^3(s) = 1 - s^2 + 2s^3 + s^4$$

$$K_2^2(s) = K_2^4(s) = 1 + s - s^2 + 2s^3 + s^4.$$

The set $\mathbf{P}_E^1(s)$ therefore consists of the 2 plant line segments

$$\frac{\lambda_1 K_1^1(s) + (1-\lambda_1)K_1^3(s)}{K_2^1(s)} : \lambda_1 \in [0, 1]$$

$$\frac{\lambda_2 K_1^1(s) + (1-\lambda_2)K_1^3(s)}{K_2^2(s)} : \lambda_2 \in [0, 1].$$

The set $\mathbf{P}_E^2(s)$ consists of the 2 plant line segments

$$\frac{K_1^1(s)}{\lambda_3 K_2^1(s) + (1-\lambda_3)K_2^2(s)} : \lambda_3 \in [0, 1]$$

$$\frac{K_1^3(s)}{\lambda_4 K_2^1(s) + (1-\lambda_4)K_2^2(s)} : \lambda_4 \in [0, 1].$$

Thus, the extremal set $\mathbf{P}_E(s)$ consists of the following four plant line segments.

$$\frac{0.5(1+\lambda_1) - 2s - (1+\lambda_1)s^2 + s^3}{1 - s^2 + 2s^3 + s^4} : \lambda_1 \in [0, 1]$$

$$\frac{0.5(1+\lambda_2) - 2s - (1+\lambda_2)s^2 + s^3}{1 + s - s^2 + 2s^3 + s^4} : \lambda_2 \in [0, 1]$$

$$\frac{0.5 - 2s - s^2 + s^3}{1 + \lambda_3 s - s^2 + 2s^3 + s^4} : \lambda_3 \in [0, 1],$$

$$\frac{1 - 2s - 2s^2 + s^3}{1 + \lambda_4 s - s^2 + 2s^3 + s^4} : \lambda_4 \in [0, 1].$$

Therefore, we can verify robust stability by checking the Hurwitz stability of the set $\Delta_E(s)$ which consists of the following four polynomial line segments.

$$F_1(s)\big(0.5(1+\lambda_1) - 2s - (1+\lambda_1)s^2 + s^3\big) + F_2(s)\big(1 - s^2 + 2s^3 + s^4\big)$$

$$F_1(s)\big(0.5(1+\lambda_2) - 2s - (1+\lambda_2)s^2 + s^3\big) + F_2(s)\big(1 + s - s^2 + 2s^3 + s^4\big)$$

$$F_1(s)\big(0.5 - 2s - s^2 + s^3\big) + F_2(s)\big(1 + \lambda_3 s - s^2 + 2s^3 + s^4\big)$$

$$F_1(s)\big(1 - 2s - 2s^2 + s^3\big) + F_2(s)\big(1 + \lambda_4 s - s^2 + 2s^3 + s^4\big)$$

for $\lambda_i \in [0, 1] : i = 1, 2, 3, 4$.

In other words, any compensator that stabilizes the family of plants $\mathbf{P}(s)$ must stabilize the 4 one-parameter family of extremal plants $\mathbf{P}_E(s)$. If we had used the Edge Theorem it would have been necessary to check the 12 line segments of plants corresponding to the exposed edges of $\Delta(s)$.

If the compensator polynomials $F_i(s)$ satisfy the "vertex conditions" in part II of GKT, it is enough to check that they stabilize the plants corresponding to the four Kharitonov vertices. This corresponds to checking the Hurwitz stability of the four *fixed* polynomials

$$F_1(s)\left(1-2s-2s^2+s^3\right)+F_2(s)\left(1-s^2+2s^3+s^4\right)$$
$$F_1(s)\left(1-2s-2s^2+s^3\right)+F_2(s)\left(1+s-s^2+2s^3+s^4\right)$$
$$F_1(s)\left(0.5-2s-s^2+s^3\right)+F_2(s)\left(1+s-s^2+2s^3+s^4\right)$$
$$F_1(s)\left(0.5-2s-s^2+s^3\right)+F_2(s)\left(1-s^2+2s^3+s^4\right).$$

Remark 6.2. The extremal set used in the GKT plays a fundamental role in frequency domain design methods and generates the frequency domain templates necessary for Nyquist and Bode analysis. For examples see Bhattacharyya et al. (1995, 2009). It is also shown in Bhattacharyya et al. (1995) that the minimum value of the parametric stability margin $\rho(p)$ occurs on the Kharitonov segments in the extremal subset. These points are thus the "weakest points" in the family of systems from the point of view of robustness.

7. Computation of the parametric stability margin

In this section we give a useful characterization of the parametric stability margin in the general case. This can be done by finding the largest stability ball in parameter space, centered at a "stable" nominal parameter value $\mathbf{p}^0$. The results to be described here were developed in Soh, Berger, and Dabke (1985), Tesi and Vicino (1989), Vicino (1991), Tsypkin and Polyak (1991), and Biernacki, Hwang, Bhattacharyya (1986). Let $\mathcal{S}\subset\mathbb{C}$ denote as usual an open set which is symmetric with respect to the real axis. $\mathcal{S}$ denotes the stability region of interest. In continuous-time systems $\mathcal{S}$ may be the open left-half plane or subsets thereof. For discrete-time systems $\mathcal{S}$ is the open unit circle or a subset of it. Now let $\mathbf{p}$ be a vector of real parameters,

$$\mathbf{p}=[p_1,p_2,\ldots,p_l]^T.$$

The characteristic polynomial of the system is denoted by

$$\delta(s,\mathbf{p})=\delta_n(\mathbf{p})s^n+\delta_{n-1}(\mathbf{p})s^{n-1}+\cdots+\delta_0(\mathbf{p}).$$

The polynomial $\delta(s,\mathbf{p})$ is a real polynomial with coefficients that depend continuously on the real parameter vector $\mathbf{p}$. We suppose that for the *nominal parameter* $\mathbf{p}=\mathbf{p}^0$, $\delta(s,\mathbf{p}^0):=\delta^0(s)$ is stable with respect to $\mathcal{S}$ (has its roots in $\mathcal{S}$). Write

$$\triangle\mathbf{p}:=\mathbf{p}-\mathbf{p}^0=[p_1-p_1^0,p_2-p_2^0,\ldots,p_l-p_l^0]$$

to denote the perturbation in the parameter $\mathbf{p}$ from its nominal value $\mathbf{p}^0$. Now let us introduce a norm $\|\cdot\|$ in the space of the parameters $\mathbf{p}$ and introduce the open ball of radius ρ

$$\mathcal{B}(\rho,\mathbf{p}^0)=\{\mathbf{p}:\|\mathbf{p}-\mathbf{p}^0\|<\rho\}. \tag{32}$$

The hypersphere of radius ρ is defined by

$$\mathcal{S}(\rho,\mathbf{p}^0)=\{\mathbf{p}:\|\mathbf{p}-\mathbf{p}^0\|=\rho\}. \tag{33}$$

With the ball $\mathcal{B}(\rho,\mathbf{p}^0)$ we associate the family of uncertain polynomials:

$$\mathbf{\Delta}_\rho(s):=\{\delta(s,\mathbf{p}^0+\Delta\mathbf{p}):\|\Delta\mathbf{p}\|<\rho\}. \tag{34}$$

Definition 7.1. The *real parametric stability margin* in parameter space is defined as the radius, measured in some norm and denoted $\rho^*(\mathbf{p}^0)$, of the *largest* ball centered at $\mathbf{p}^0$ for which $\delta(s,\mathbf{p})$ remains stable whenever $\mathbf{p}\in\mathcal{B}(\rho^*(\mathbf{p}^0),\mathbf{p}^0)$.

This stability margin then tells us how much we can perturb the original parameter $\mathbf{p}^0$ and yet remain stable. Our first result is a characterization of this maximal stability ball. To simplify notation we write ρ^* instead of $\rho^*(\mathbf{p}^0)$.

Theorem 7.2. *Bhattacharyya et al. (1995) With the assumptions as above the parametric stability margin ρ^* is characterized by:*

- *(a) There exists a largest stability ball $\mathcal{B}(\rho^*,\mathbf{p}^0)$ centered at $\mathbf{p}^0$, with the property that:*
  - *(a1) For every $\mathbf{p}'$ within the ball, the characteristic polynomial $\delta(s,\mathbf{p}')$ is stable and of degree n.*
  - *(a2) At least one point $\mathbf{p}''$ on the hypersphere $\mathcal{S}(\rho^*,\mathbf{p}^0)$ itself is such that $\delta(s,\mathbf{p}'')$ is unstable or of degree less than n.*
- *(b) Moreover if $\mathbf{p}''$ is any point on the hypersphere $\mathcal{S}(\rho^*,\mathbf{p}^0)$ such that $\delta(s,\mathbf{p}'')$ is unstable, then the unstable roots of $\delta(s,\mathbf{p}'')$ can only be on the stability boundary.*

The parametric stability margin or distance to instability is measured in the norm $\|\cdot\|$, and therefore the numerical value of ρ^* will depend on the specific norm chosen. We will consider, in particular, the weighted ℓ_p norms. These are defined as follows: Let $w=[w_1,w_2,\cdots,w_l]$ with $w_i>0$ be a set of positive *weights*.

$$\ell_1 \ \text{norm} \ : \quad \|\Delta\mathbf{p}\|_1^w:=\sum_{i=1}^{l} w_i|\Delta p_i|$$

$$\ell_2 \ \text{norm} \ : \quad \|\Delta\mathbf{p}\|_2^w:=\sqrt{\sum_{i=1}^{l} w_i^2\Delta p_i^2}$$

$$\ell_p \ \text{norm} \ : \quad \|\Delta\mathbf{p}\|_p^w:=\left[\sum_{i=1}^{l}|w_i\Delta p_i|^p\right]^{\frac{1}{p}}$$

$$\ell_\infty \ \text{norm} \ : \quad \|\Delta\mathbf{p}\|_\infty^w:=\max_i w_i|\Delta p_i|.$$

We will write $\|\triangle\mathbf{p}\|$ to refer to a generic weighted norm when the weight and type of norm are unimportant.

7.1. The image set approach

The parametric stability margin may be calculated by using the complex plane image of the polynomial family $\mathbf{\Delta}_\rho(s)$ evaluated at each point on the stability boundary $\partial\mathcal{S}$. This is based on the following idea. Suppose that the family has constant degree n and $\delta(s,\mathbf{p}^0)$ is stable but $\mathbf{\Delta}_\rho(s)$ contains an unstable polynomial. Then the continuous dependence of the roots on $\mathbf{p}$ and the Boundary Crossing Theorem imply that there must also exist a polynomial in $\mathbf{\Delta}_\rho(s)$ such that it has a root, at a point, s^*, on the stability boundary $\partial\mathcal{S}$. In this case *the complex plane image set $\mathbf{\Delta}_\rho(s^*)$ must contain the origin of the complex plane*. This suggests that to detect the presence of instability in a family of polynomials of constant degree, we generate the image set of the family at each point of the stability boundary and determine if the origin is included in or excluded from this set.

Theorem 7.3 (Zero Exclusion Principle). *Bhattacharyya et al. (1995) For given $\rho\ge 0$ and $\mathbf{p}^0$, suppose that the family of polynomials $\mathbf{\Delta}_\rho(s)$ is of constant degree and $\delta(s,\mathbf{p}^0)$ is stable. Then every polynomial in the family $\mathbf{\Delta}_\rho(s)$ is stable with respect to $\mathcal{S}$ if and only if the complex plane image set $\mathbf{\Delta}_\rho(s^*)$ excludes the origin for every $s^*\in\partial\mathcal{S}$.*

In fact, the above can be used as a computational tool to determine the maximum value ρ^* of ρ for which the family is stable. If $\delta(s,\mathbf{p}^0)$ is stable, it follows that there always exists an open stability ball around $\mathbf{p}^0$ since the stability region $\mathcal{S}$ is itself open. Therefore, for small values of ρ the image set $\mathbf{\Delta}_\rho(s^*)$ will exclude the origin for every point $s^*\in\partial\mathcal{S}$. As ρ is increased from zero, a *limiting* value ρ_0 may be reached where some polynomial in the corresponding family $\mathbf{\Delta}_{\rho_0}(s)$ loses degree or a polynomial in the family acquires a root s^* on the stability boundary. From Theorem 7.2 it is clear that this value ρ_0 is equal to ρ^*, the stability margin. In case the limiting value ρ_0 is never achieved, the stability margin ρ^* is infinity.

An alternative way to determine ρ^* is as follows: Fixing s^* at a point in the boundary of $\mathcal{S}$, let $\rho_0(s^*)$ denote the limiting value of

ρ such that $0 \in \mathbf{\Delta}_\rho(s^*)$:

$$\rho_0(s^*) := \inf\{\rho : 0 \in \mathbf{\Delta}_\rho(s^*)\}.$$

Then, we define

$$\rho_b = \inf_{s^* \in \partial\mathcal{S}} \rho_0(s^*).$$

In other words, ρ_b is the limiting value of ρ for which some polynomial in the family $\mathbf{\Delta}_\rho(s)$ *acquires a root on the stability boundary* $\partial\mathcal{S}$. Also let the limiting value of ρ for which some polynomial in $\mathbf{\Delta}_\rho(s)$ *loses degree* be denoted by ρ_d:

$$\rho_d := \inf\left\{\rho : \delta_n(\mathbf{p}^0 + \Delta\mathbf{p}) = 0, \quad \|\Delta\mathbf{p}\| < \rho\right\}.$$

We can now state the following theorem.

Theorem 7.4. Bhattacharyya et al. (1995) *The parametric stability margin*

$$\rho^* = \min\{\rho_b, \rho_d\}.$$

Remark 7.5. We note that this theorem remains valid even when the stability region $\mathcal{S}$ is not connected. For instance, one may construct the stability region as a union of disconnected regions $\mathcal{S}_i$ surrounding each root of the nominal polynomial. In this case the stability boundary must also consist of the union of the individual boundary components $\partial\mathcal{S}_i$. The functional dependence of the coefficients δ_i on the parameters $\mathbf{p}$ is also not restricted in any way except for the assumption of continuity.

The above theorem shows that the problem of determining ρ^* can be reduced to the following steps:

(A) determine the "local" stability margin $\rho(s^*)$ at each point s^* on the boundary of the stability region,
(B) minimize the function $\rho(s^*)$ over the entire stability boundary and thus determine ρ_b,
(C) calculate ρ_d, and set,
(D) $\rho^* = \min\{\rho_b, \rho_d\}$.

In general the determination of ρ^* is a *difficult nonlinear optimization* problem. However, the breakdown of the problem into the steps described above, exploits the structure of the problem and has the advantage that the local stability margin calculation $\rho(s^*)$ with s^* frozen, can be simple. In particular, when the parameters enter linearly into the characteristic polynomial coefficients, this calculation can be done in closed form. It reduces to a least square problem for the ℓ_2 case, and equally simple linear programming or vertex problems for the ℓ_∞ or ℓ_1 cases. The dependence of ρ_b on s^* is in general highly nonlinear but this part of the minimization can be easily performed computationally because sweeping the boundary $\partial\mathcal{S}$ is a *one-dimensional search*. In the next section, we describe and develop this calculation in greater detail.

7.2. Stability margin computation

We develop explicit formulas for the parametric stability margin in the case in which the characteristic polynomial coefficients depend linearly on the uncertain parameters. In such cases we may write without loss of generality that

$$\delta(s, \mathbf{p}) = a_1(s)p_1 + \cdots + a_l(s)p_l + b(s) \tag{35}$$

where $a_i(s)$ and $b(s)$ are real polynomials and the parameters p_i are real. As before, we write $\mathbf{p}$ for the vector of uncertain parameters, $\mathbf{p}^0$ denotes the nominal parameter vector and $\triangle\mathbf{p}$ the perturbation vector. In other words

$$\mathbf{p} = [p_1, p_2, \ldots, p_l] \qquad \mathbf{p}^0 = [p_1^0, p_2^0, \ldots, p_l^0]$$
$$\begin{aligned}\triangle\mathbf{p} &= [p_1 - p_1^0, p_2 - p_2^0, \ldots, p_l - p_l^0]\\ &= [\triangle p_1, \triangle p_2, \ldots, \triangle p_l].\end{aligned}$$

Then the characteristic polynomial can be written as

$$\delta(s, \mathbf{p}^0 + \Delta\mathbf{p}) = \underbrace{\delta(s, \mathbf{p}^0)}_{\delta^0(s)} + \underbrace{a_1(s)\triangle p_1 + \cdots + a_l(s)\triangle p_l}_{\triangle\delta(s,\triangle\mathbf{p})}. \tag{36}$$

Now let s^* denote a point on the stability boundary $\partial\mathcal{S}$. For $s^* \in \partial\mathcal{S}$ to be a root of $\delta(s, \mathbf{p}^0 + \triangle\mathbf{p})$ we must have

$$\delta(s^*, \mathbf{p}^0) + a_1(s^*)\triangle p_1 + \cdots + a_l(s^*)\triangle p_l = 0. \tag{37}$$

We rewrite this equation introducing the weights $w_i > 0$.

$$\delta(s^*, \mathbf{p}^0) + \frac{a_1(s^*)}{w_1} w_1\triangle p_1 + \cdots + \frac{a_l(s^*)}{w_l} w_l\triangle p_l = 0. \tag{38}$$

Obviously, the minimum $\|\triangle\mathbf{p}\|^w$ norm solution of this equation gives us $\rho(s^*)$, the calculation involved in step A in the last section.

$$\rho(s^*) = \inf\left\{\|\triangle\mathbf{p}\|^w \ : \ \delta(s^*, \mathbf{p}^0) + \frac{a_1(s^*)}{w_1} w_1\triangle p_1 + \cdots + \frac{a_l(s^*)}{w_l} w_l\triangle p_l = 0\right\}.$$

Similarly, corresponding to loss of degree we have the equation

$$\delta_n(\mathbf{p}^0 + \Delta\mathbf{p}) = 0. \tag{39}$$

Letting a_{in} denote the coefficient of the nth degree term in the polynomial $a_i(s)$, $i = 1, 2, \ldots, l$ the above equation becomes

$$\underbrace{a_{1n}p_1^0 + a_{2n}p_2^0 + \cdots + a_{ln}p_l^0}_{\delta_n(\mathbf{p}^0)} + a_{1n}\triangle p_1 + a_{2n}\triangle p_2 + \cdots a_{ln}\triangle p_l = 0. \tag{40}$$

We can rewrite this after introducing the weight $w_i > 0$

$$\underbrace{a_{1n}p_1^0 + a_{2n}p_2^0 + \cdots + a_{ln}p_l^0}_{\delta_n(\mathbf{p}^0)} + \frac{a_{1n}}{w_1} w_1\triangle p_1 + \frac{a_{2n}}{w_2} w_2\triangle p_2 + \cdots + \frac{a_{ln}}{w_l} w_l\triangle p_l = 0. \tag{41}$$

The minimum norm $\|\triangle\mathbf{p}\|^w$ solution of this equation gives us ρ_d.

We consider the above equations in some more detail. Recall that the polynomials are assumed to be real. Eq. (41) is real and can be rewritten in the form

$$\underbrace{\begin{bmatrix} \frac{a_{1n}}{w_1} & \cdots & \frac{a_{ln}}{w_l} \end{bmatrix}}_{A_n} \underbrace{\begin{bmatrix} w_1\triangle p_1 \\ \vdots \\ w_l\triangle p_l \end{bmatrix}}_{t_n} = \underbrace{-\delta_n^0}_{b_n}. \tag{42}$$

In (38), two cases may occur depending on whether s^* is real or complex. If $s^* = s_r$ where s_r is real, we have the single equation

$$\underbrace{\begin{bmatrix} \frac{a_1(s_r)}{w_1} & \cdots & \frac{a_l(s_r)}{w_l} \end{bmatrix}}_{A(s_r)} \underbrace{\begin{bmatrix} w_1\triangle p_1 \\ \vdots \\ w_l\triangle p_l \end{bmatrix}}_{t(s_r)} = \underbrace{-\delta^0(s_r)}_{b(s_r)}. \tag{43}$$

Let x_r and x_i denote the real and imaginary parts of a complex number x, i.e.

$$x = x_r + jx_i \qquad \text{with } x_r, x_i \text{ real.}$$

Using this notation, we will write

$$a_k(s^*) = a_{kr}(s^*) + ja_{ki}(s^*)$$

and

$$\delta^0(s^*) = \delta_r^0(s^*) + j\delta_i^0(s^*).$$

If $s^* = s_c$ where s_c is complex, (38) is equivalent to two equations which can be written as follows:

$$\underbrace{\begin{bmatrix} \frac{a_{1r}(s_c)}{w_1} & \dots & \frac{a_{lr}(s_c)}{w_l} \\ \frac{a_{1i}(s_c)}{w_1} & \dots & \frac{a_{li}(s_c)}{w_l} \end{bmatrix}}_{A(s_c)} \underbrace{\begin{bmatrix} w_1 \triangle p_1 \\ \vdots \\ w_l \triangle p_l \end{bmatrix}}_{t(s_c)} = \underbrace{\begin{bmatrix} -\delta_r^0(s_c) \\ -\delta_i^0(s_c) \end{bmatrix}}_{b(s_c)}. \tag{44}$$

These equations completely determine the parametric stability margin in any norm. Let $t^*(s_c)$, $t^*(s_r)$, and t_n^* denote the minimum norm solutions of (44), (43), and (42), respectively. Thus,

$$\|t^*(s_c)\| = \rho(s_c) \tag{45}$$

$$\|t^*(s_r)\| = \rho(s_r) \tag{46}$$

$$\|t_n^*\| = \rho_d. \tag{47}$$

If any of the above Eqs. (42)–(44) do not have a solution, the corresponding value of $\rho(\,\cdot\,)$ is set equal to infinity.

Let $\partial\mathcal{S}_r$ and $\partial\mathcal{S}_c$ denote the real and complex subsets of $\partial\mathcal{S}$:

$$\partial\mathcal{S} = \partial\mathcal{S}_r \cup \partial\mathcal{S}_c.$$

$$\rho_r := \inf_{s_r \in \partial\mathcal{S}_r} \rho(s_r)$$

$$\rho_c := \inf_{s_c \in \partial\mathcal{S}_c} \rho(s_c),$$

Therefore,

$$\rho_b = \inf\{\rho_r, \rho_c\}. \tag{48}$$

We now consider the specific case of the ℓ_2 norm.

7.3. ℓ_2 stability margin

In this section we suppose that the length of the perturbation vector $\triangle\mathbf{p}$ is measured by a weighted ℓ_2 norm. That is, the minimum ℓ_2 norm solution of Eqs. (42)–(44) are desired. Consider first (44). Assuming that $A(s_c)$ has full row rank=2, the minimum norm solution vector $t^*(s_c)$ can be calculated as follows:

$$t^*(s_c) = A^T(s_c)\left[A(s_c)A^T(s_c)\right]^{-1} b(s_c). \tag{49}$$

Similarly if (43) and (42) are consistent (i.e., $A(s_r)$ and A_n are nonzero vectors), we can calculate the solution as

$$t^*(s_r) = A^T(s_r)\left[A(s_r)A^T(s_r)\right]^{-1} b(s_r) \tag{50}$$

$$t_n^* = A_n^T\left[A_n A_n^T\right]^{-1} b_n. \tag{51}$$

If $A(s_c)$ has less than full rank the following two cases can occur.

Case 1: Rank $A(s_c) = 0$ In this case the equation is inconsistent since $b(s_c) \neq 0$ (otherwise $\delta^0(s_c) = 0$ and $\delta^0(s)$ would not be stable with respect to $\mathcal{S}$ since s_c lies on $\partial\mathcal{S}$). In this case (44) has no solution and we set

$$\rho(s_c) = \infty.$$

Case 2: Rank $A(s_c) = 1$ In this case the equation is consistent if and only if

$$\text{rank}[A(s_c), b(s_c)] = 1.$$

Example 7.6 (ℓ_2 Schur stability margin)**.** Consider the discrete time control system with the controller and plant specified respectively by their transfer functions:

$$C(z) = \frac{z+1}{z^2}, \quad G(z, \mathbf{p}) = \frac{(-0.5 - 2p_0)z + (0.1 + p_0)}{z^2 - (1 + 0.4p_2)z + (0.6 + 10p_1 + 2p_0)}.$$

The characteristic polynomial of the closed-loop system is:

$$\delta(z, \mathbf{p}) = z^4 - (1 + 0.4p_2)z^3 + (0.1 + 10p_1)z^2 - (0.4 + p_0)z + (0.1 + p_0).$$

The nominal value of $\mathbf{p}^0 = [p_0^0 \ p_1^0 \ p_2^0] = [0,\ 0.1,\ 1]$. The perturbation is denoted as usual by the vector

$$\Delta\mathbf{p} = \begin{bmatrix} \Delta p_0 & \Delta p_1 & \Delta p_2 \end{bmatrix}.$$

The polynomial is Schur stable for the nominal parameter $\mathbf{p}^0$. We compute the ℓ_2 stability margin of the polynomial with weights $w_1 = w_2 = w_3 = 1$. Rewrite

$$\delta(z, \mathbf{p}^0 + \Delta\mathbf{p}) = (-z+1)\Delta p_0 + 10z^2\Delta p_1 - 0.4z^3\Delta p_2 + (z^4 - 1.4z^3 + 1.1z^2 - 0.4z + 0.1)$$

and note that the degree remains invariant (=4) for all perturbations so that $\rho_d = \infty$. The stability region is the unit circle. For $z = 1$ to be a root of $\delta(z, \mathbf{p}^0 + \Delta\mathbf{p})$ (see (43)), we have

$$\underbrace{\begin{bmatrix} 0 & 10 & -0.4 \end{bmatrix}}_{A(1)} \underbrace{\begin{bmatrix} \Delta p_0 \\ \Delta p_1 \\ \Delta p_2 \end{bmatrix}}_{t(1)} = \underbrace{-0.4}_{b(1)}.$$

Thus,

$$\rho(1) = \|t^*(1)\|_2 = \left\| A^T(1)\left[A(1)A^T(1)\right]^{-1} b(1) \right\|_2 = 0.04\ .$$

Similarly, for the case of $z = -1$ (see (43)), we have

$$\underbrace{\begin{bmatrix} 2 & 10 & 0.4 \end{bmatrix}}_{A(-1)} \underbrace{\begin{bmatrix} \Delta p_0 \\ \Delta p_1 \\ \Delta p_2 \end{bmatrix}}_{t(-1)} = \underbrace{-4}_{b(-1)}.$$

and $\rho(-1) = \|t^*(-1)\|_2 = 0.3919$. Thus $\rho_r = 0.04$.

For the case in which $\delta(z, \mathbf{p}^0 + \Delta\mathbf{p})$ has a root at $z = e^{j\theta}$, $\theta \neq \pi$, $\theta \neq 0$, using (44), we have

$$\underbrace{\begin{bmatrix} -\cos\theta + 1 & 10\cos 2\theta & -0.4\cos 3\theta \\ -\sin\theta & 10\sin 2\theta & -0.4\sin 3\theta \end{bmatrix}}_{A(\theta)} \underbrace{\begin{bmatrix} \Delta p_0 \\ \Delta p_1 \\ \Delta p_2 \end{bmatrix}}_{t(\theta)}$$

$$= \underbrace{-\begin{bmatrix} \cos 4\theta - 1.4\cos 3\theta + 1.1\cos 2\theta - 0.4\cos\theta + 0.1 \\ \sin 4\theta - 1.4\sin 3\theta + 1.1\sin 2\theta - 0.4\sin\theta \end{bmatrix}}_{b(\theta)}.$$

Thus,

$$\rho(e^{j\theta}) = \|t^*(\theta)\|_2 = \left\| A^T(\theta)\left[A(\theta)A^T(\theta)\right]^{-1} b(\theta) \right\|.$$

Fig. 7 shows the plot of $\rho(e^{j\theta})$. Therefore, the ℓ_2 parametric stability margin is

$$\rho_c = 0.032 = \rho_b = \rho^*.$$

Example 7.7. Consider the continuous time control system with the plant

$$G(s, \mathbf{p}) = \frac{2s + 3 - \frac{1}{3}p_1 - \frac{5}{3}p_2}{s^3 + (4 - p_2)s^2 + (-2 - 2p_1)s + (-9 + \frac{5}{3}p_1 + \frac{16}{3}p_2)}$$

and the proportional integral (PI) controller

$$C(s) = 5 + \frac{3}{s}.$$

The characteristic polynomial of the closed-loop system is

$$\delta(s, \mathbf{p}) = s^4 + (4 - p_2)s^3 + (8 - 2p_1)s^2 + (12 - 3p_2)s + (9 - p_1 - 5p_2).$$

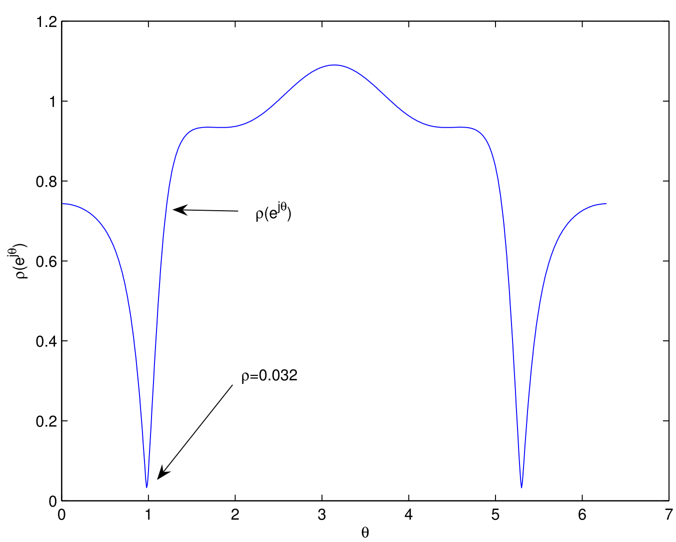

Fig. 7. $\rho(\theta)$ (Example 7.6).

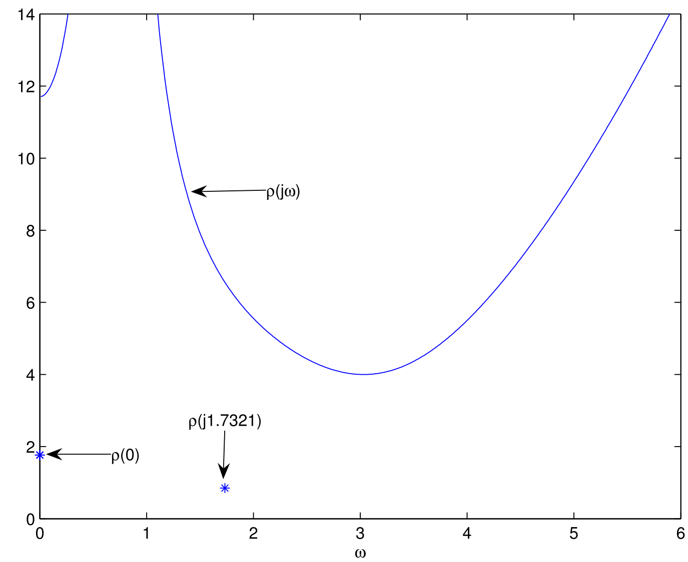

Fig. 8. $\rho(j\omega)$, (Example 7.7).

We see that the degree remains invariant under the given set of parameter variations and therefore $\rho_d = \infty$. The nominal values of the parameters are

$$\mathbf{p}^0 = [p_1^0 \ , \ p_2^0] = [0 \ , \ 0].$$

Then

$$\Delta\mathbf{p} = \begin{bmatrix}\Delta p_1 & \Delta p_2\end{bmatrix} = \begin{bmatrix}p_1 & p_2\end{bmatrix}.$$

The polynomial is stable for the nominal parameter $\mathbf{p}^0$. Now we want to compute the ℓ_2 stability margin of this polynomial with weights $w_1 = w_2 = 1$. We first evaluate $\delta(s, \mathbf{p})$ at $s = j\omega$:
$\delta(j\omega, \mathbf{p}^0 + \Delta\mathbf{p}) = (2\omega^2 - 1)\Delta p_1 + (j\omega^3 - 3j\omega - 5)\Delta p_2 + \omega^4 - 4j\omega^3 - 8\omega^2 + 12j\omega + 9.$

For the case of a root at $s = 0$ (see (43)), we have

$$\underbrace{\begin{bmatrix}-1 & -5\end{bmatrix}}_{A(0)}\underbrace{\begin{bmatrix}\Delta p_1 \\ \Delta p_2\end{bmatrix}}_{t(0)} = \underbrace{-9.}_{b(0)}$$

Thus,

$$\rho(0) = \|t^*(0)\|_2 = \left\|A^T(0)\left[A(0)A^T(0)\right]^{-1}b(0)\right\|_2 = \frac{9\sqrt{26}}{26}.$$

For a root at $s = j\omega$, $\omega > 0$, using the formula given in (44), we have with $w_1 = w_2 = 1$

$$\underbrace{\begin{bmatrix}(2\omega^2-1) & -5 \\ 0 & (\omega^2-3)\end{bmatrix}}_{A(j\omega)}\underbrace{\begin{bmatrix}\Delta p_1 \\ \Delta p_2\end{bmatrix}}_{t(j\omega)} = \underbrace{\begin{bmatrix}-\omega^4 + 8\omega^2 - 9 \\ 4\omega^2 - 12\end{bmatrix}}_{b(j\omega)}. \tag{52}$$

Here, we need to determine if there exists any ω for which the rank of the matrix $A(j\omega)$ drops. It is easy to see from the matrix $A(j\omega)$, that the rank drops when $\omega = \frac{1}{\sqrt{2}}$ and $\omega = \sqrt{3}$.

rank $A(j\omega) = 1$: For $\omega = \frac{1}{\sqrt{2}}$, we have rank $A(j\omega) = 1$ and rank$[A(j\omega), b(j\omega)] = 2$, and so there is no solution to (52). Thus,

$$\rho\left(j\frac{1}{\sqrt{2}}\right) = \infty.$$

For $\omega = \sqrt{3}$, rank $A(j\omega) = $ rank$[A(j\omega), b(j\omega)]$, and we do have a solution to (52). Therefore

$$\underbrace{\begin{bmatrix}5 & -5\end{bmatrix}}_{A(j\sqrt{3})}\underbrace{\begin{bmatrix}\Delta p_1 \\ \Delta p_2\end{bmatrix}}_{t(j\sqrt{3})} = \underbrace{6}_{b(j\sqrt{3})}.$$

Consequently,

$$\rho(j\sqrt{3}) = \left\|t^*(j\sqrt{3})\right\|_2 = \left\|A^T(j\sqrt{3})\left[A(j\sqrt{3})A^T(j\sqrt{3})\right]^{-1}b(j\sqrt{3})\right\|_2 = \frac{3\sqrt{2}}{5}.$$

rank $A(j\omega) = 2$: In this case (52) has a solution (which happens to be unique) and the length of the least square solution is found by

$$\rho(j\omega) = \|t^*(j\omega)\|_2 = \left\|A^T(j\omega)\left[A(j\omega)A^T(j\omega)\right]^{-1}b(j\omega)\right\|_2 = \sqrt{\left(\frac{\omega^4 - 8\omega^2 - 11}{2\omega^2 - 1}\right)^2 + 16}.$$

Fig. 8 shows the plot of $\rho(j\omega)$ for $\omega > 0$. The values of $\rho(0)$ and $\rho(j\sqrt{3})$ are also shown in Fig. 8. Therefore,

$$\rho(j\sqrt{3}) = \rho_b = \frac{3\sqrt{2}}{5} = \rho^*$$

is the stability margin.

7.4. ℓ_∞ and ℓ_1 stability margins

If $\Delta\mathbf{p}$ is measured in the ℓ_∞ or ℓ_1 norm, we face the problem of determining the corresponding minimum norm solution of a linear equation of the type $At = b$ at each point on the stability boundary. Problems of this type can always be converted to a suitable linear programming problem. For instance, in the ℓ_∞ case, this problem is equivalent to the following optimization problem:

Minimize β

subject to the constraints:

$At = b$

$$-\beta \le t_i \le \beta, \qquad i = 1, 2, \dots, l. \tag{53}$$

This is a standard linear programming (LP) problem and can therefore be solved by existing, efficient algorithms.

For the ℓ_1 case, we can similarly formulate a linear programming problem by introducing the variables,

$$|t_i| := y_i^+ - y_i^-, \qquad i = 1, 2, \dots, l$$

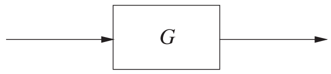

Fig. 9. An open loop system.

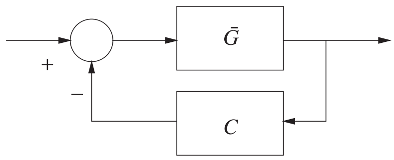

Fig. 10. A feedback system.

Then we have the LP problem:

$$\text{Minimize} \sum_{i=1}^{l} \{y_i^+ - y_i^-\} \tag{54}$$

subject to

$$At = b$$

$$y_i^- - y_i^+ \le t_i \le y_i^+ - y_i^- \qquad i = 1, 2, \ldots, l.$$

We do not elaborate further on this approach. The reason is that ℓ_∞ and ℓ_1 cases are special cases of polytopic families of perturbations, which were dealt with earlier. A direct approach to these calculations is with the Tsypkin–Polyak locus described in Tsypkin and Polyak (1991). We have dealt with the case where the parameters appear affinely in the polynomial coefficients, where they appear multilinearly in the characteristic polynomial coefficients, the Mapping Theorem Zadeh and Deoser (1976) may be used to obtain "estimates" of the margins (see Bhattacharyya et al. (1995, 2009)).

8. Controller fragility of high order controllers

In this section we focus on the 1997 results of Keel and Bhattacharyya (1997), where it was shown that high order controllers, even those designed to be robust to plant uncertainty, namely plant-robust, could be very fragile with respect to controller parameters, that is controller fragile.

8.1. Robustness using high gain feedback

In this section we illustrate how robust systems can be built from highly unreliable components by using high gain and the feedback structure. These ideas were introduced by H.S. Black in 1926 when he invented the feedback amplifier (Kline, 1993). Specifically we consider the problem of obtaining a precisely controlled value of gain, from a system containing large parameter uncertainty. Consider the system shown in Fig. 9.

Suppose that the system gain G is required to be 100. Due to poor reliability of the components, this can vary by 50%. Then the actual gain can range from 50 to 150. To remedy this situation, we use the feedback structure shown in Fig. 10.

The gain $\bar{G}$ is again made with the same unreliable components but with nominal value much higher than 100, say 10,000, and we set C = 0.01. The overall gain of the feedback system is given by the expression

$$\frac{\bar{G}}{1 + C\bar{G}}. \tag{55}$$

With 50% variation in $\bar{G}$ (that is, $\bar{G}$ varies from 5000 to 15,000) the gain of the feedback system varies from 98.039 to 99.338. This remarkable increase in robustness, corresponding to a reduction of uncertainty from 50% to 1%, is one of the main reasons for the widespread use of feedback in the control and electronics industry.

As a final observation it is not difficult to see that robustness to variations in G is obtained at the cost of *fragility* with respect to C.

8.2. Fragility of high order controllers

The purpose of this section is to analyze some high order controller designs from the published literature and examine the robustness of the resulting feedback systems. We analyze three examples of optimal designs taken from the control literature. Each example is a single-input single-output system and the controller is designed to optimize some closed loop performance function. For more details, the readers should refer to Keel and Bhattacharyya (1997).

Example 8.1 (H_∞ based optimum gain margin controller)**.** This example is found in (Keel and Bhattacharyya (1997), p. 200). It uses the YJBK parametrization and the machinery of the H_∞ Model Matching Problem to optimize the upper gain margin. The plant to be controlled is

$$P(s) = \frac{s-1}{s^2 - s - 2}$$

and the controller, designed to give an upper gain margin of 3.5, (the closed loop is stable for the gain interval [1, 3.5]) is obtained by optimizing the H_∞ norm of a complementary sensitivity function. The controller found is

$$C(s) = \frac{q_6^0 s^6 + q_5^0 s^5 + q_4^0 s^4 + q_3^0 s^3 + q_2^0 s^2 + q_1^0 s + q_0^0}{p_6^0 s^6 + p_5^0 s^5 + p_4^0 s^4 + p_3^0 s^3 + p_2^0 s^2 + p_1^0 s + p_0^0}$$

where

$$\begin{aligned} q_6^0 &= 379 & p_6^0 &= 3 \\ q_5^0 &= 39383 & p_5^0 &= -328 \\ q_4^0 &= 192306 & p_4^0 &= -38048 \\ q_3^0 &= 382993 & p_3^0 &= -179760 \\ q_2^0 &= 383284 & p_2^0 &= -314330 \\ q_1^0 &= 192175 & p_1^0 &= -239911 \\ q_0^0 &= 38582 & p_0^0 &= -67626. \end{aligned}$$

The poles of this nominal controller are:

$$174.70, \quad -65.99, \quad -1.86, \quad -1.04, \quad -0.98 \pm j0.03$$

and the poles of the closed loop system are:

$$\begin{aligned} &-0.4666 \pm j14.2299, \quad -5.5334 \pm j11.3290, \quad -1.0002, \\ &-1.0000 \pm j0.0002, \quad -0.9998 \end{aligned}$$

and this verifies that the controller is indeed stabilizing.

The Nyquist plot of $P(s)C(s)$ is shown in Fig. 11 and verifies that the desired upper gain margin is achieved. On the other hand, we see from Fig. 11 that the lower gain margin and phase margin are:

$$\text{Gain Margin} = [1, 0.9992]$$

$$\text{Phase Margin} = [0, 0.1681] \text{ degree}$$

This means roughly that a reduction in gain of one part in one thousand will destabilize the closed loop system! Likewise a vanishingly small phase perturbation is destabilizing.

To continue with our analysis let us consider the transfer function coefficients of the controller to be a parameter vector $\mathbf{p}$ with its nominal value being

$$\mathbf{p}^0 = \left[q_6^0 \ \cdots \ q_0^0 \ p_6^0 \ \cdots p_0^0 \right]$$

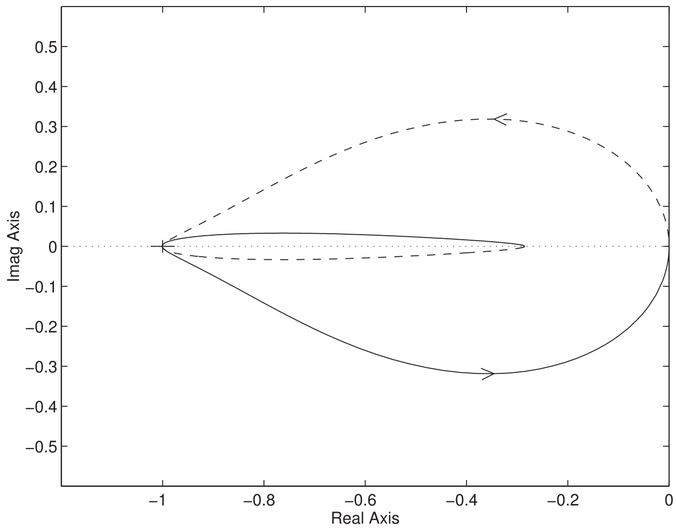

Fig. 11. Nyquist plot of $P(s)C(s)$.

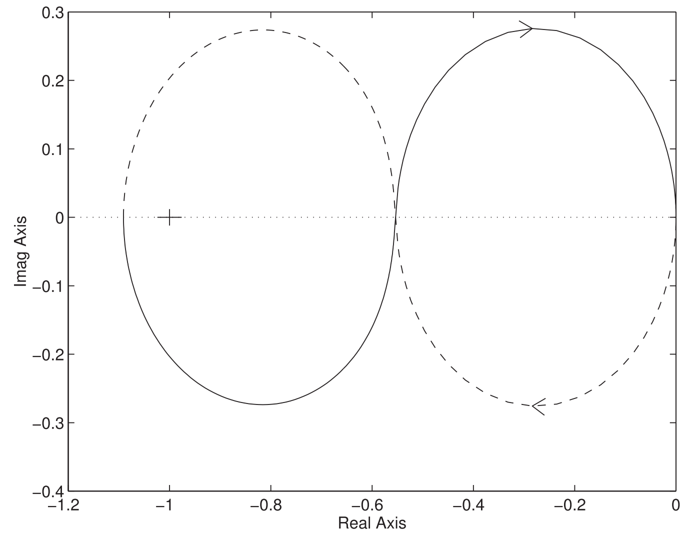

Fig. 12. Nyquist plot of $P(s)C(s)$.

and let $\Delta\mathbf{p}$ be the vector representing perturbations in $\mathbf{p}$. We compute the l^2 parametric stability margin around the nominal point. This comes out to be:

$$\rho = 0.15813903109631.$$

The normalized ratio of change in controller coefficients required to destabilize the closed loop is

$$\frac{\rho}{\|\mathbf{p}^0\|_2} = 2.103407115900516 \times 10^{-7}.$$

This shows that a change in the controller coefficients of less than 1 part in a million destabilizes the closed loop. This controller is anything but robust; in fact we are certainly justified in labeling it as a *fragile* controller.

In order to verify this rather surprising result, we construct the destabilizing controller whose parameters are obtained by setting $\mathbf{p} = \mathbf{p}^0 + \Delta\mathbf{p}$ and are:

$$\begin{aligned}
q_6 &= 379.000285811, & p_6 &= 3.158134748\\
q_5 &= 39382.999231141, & p_5 &= -327.999718909\\
q_4 &= 192305.999998597, & p_4 &= -38048.000776386\\
q_3 &= 382993.000003775, & p_3 &= -179760.000001380\\
q_2 &= 383284.000000007, & p_2 &= -314329.999996188\\
q_1 &= 192174.999999982, & p_1 &= -239910.999999993\\
q_0 &= 38582.000000000, & p_0 &= -67626.000000018
\end{aligned}$$

The closed loop poles of the system with this controller are:

$$0.000 \pm j14.2717, \quad -5.5745 \pm j10.9187,$$
$$-1.0067 \pm j\ 0.0158, \quad -1.0044, \quad -0.9820$$

which shows that the roots crossover to the right-half plane at $\omega = 14.27$, and the perturbed controller is indeed destabilizing.

Example 8.2 (H_∞ robust controller)**.** The following example is also taken from (Keel and Bhattacharyya, 1997, p.192). This example designs an optimal H_∞ robust controller that minimizes $\|W_2(s)T(s)\|_\infty$ where $T(s)$ is the complementary sensitivity function and the weight $W_2(s)$ is chosen as the high-pass function

$$W_2(s) = \frac{s+0.1}{s+1}.$$

The plant transfer function is

$$P(s) = \frac{s-1}{s^2+0.5s-0.5}.$$

and the optimally robust controller found is

$$C(s) = \frac{-124.5s^3 - 364.95s^2 - 360.45s - 120}{s^3 + 227.1s^2 + 440.7s + 220}.$$

The poles of the closed loop system are:

$$\begin{aligned}
&-99.99999999999994\\
&-1.00000484275516\\
&-0.99999757862242 \pm j\ 0.00000419348343\\
&-0.10000000000000
\end{aligned}$$

and therefore the controller does stabilize the nominal plant.

For the purposes of our analysis we first took the controller coefficients as a parameter vector $\mathbf{p}$ and found that the parametric stability margin (l^2 norm of the smallest destabilizing perturbation $\Delta\mathbf{p}$) to be:

$$\rho = 8.94427190999916.$$

The normalized ratio of change in controller coefficients required for destabilization is:

$$\frac{\rho}{\|\mathbf{p}^0\|_2} = 0.01167214151733$$

which shows that the controller, which by design is maximally robust with respect to H_∞ perturbations, is quite fragile with respect to controller coefficient perturbations.

To continue our analysis the Nyquist plot of the system with this controller is drawn in Fig. 12. From this we obtain lower gain and phase margin:

$$\text{Gain Margin} = [1, 0.91666666666667]$$
$$\text{Phase Margin} = [0, 12.91170327761722] \text{ degrees}$$

which are quite poor and would probably be unacceptable in a real system.

Example 8.3 (μ based design)**.** In this example Bhattacharyya et al. (1995) a robust controller for an electromagnetic suspension system is designed by using the μ-synthesis technique.

The plant transfer function is

$$P(s) = \frac{-36.27}{s^3 + 45.69s^2 - 4480.9636s - 204735.226884}$$

and the controller designed to tolerate prescribed structured plant perturbations is given as

$$C(s) = \frac{q_6^0 s^6 + q_5^0 s^5 + q_4^0 s^4 + q_3^0 s^3 + q_2^0 s^2 + q_1^0 s + q_0^0}{s^7 + p_6^0 s^6 + p_5^0 s^5 + p_4^0 s^4 + p_3^0 s^3 + p_2^0 s^2 + p_1^0 s + p_0^0}$$

where

$q_6^0 = -5.2200000000 \times 10^8, \quad p_6^0 = 1.4681700000 \times 10^3$

$q_5^0 = -1.1906298000 \times 10^{11}, \quad p_5^0 = 8.1539147240 \times 10^5$

$q_4^0 = -1.0892119024 \times 10^{13}, \quad p_4^0 = 2.2686802480 \times 10^8$

$q_3^0 = -5.1046222520 \times 10^{14}, \quad p_3^0 = 1.8187634284 \times 10^{10}$

$q_2^0 = -1.2852702618 \times 10^{16}, \quad p_2^0 = 5.6984090389 \times 10^{11}$

$q_1^0 = -1.6295326897 \times 10^{17}, \quad p_1^0 = 6.2845429258 \times 10^{12}$

$q_0^0 = -7.9372179723 \times 10^{17}, \quad p_0^0 = 6.2277404850 \times 10^{11}.$

The poles of the closed loop system are:

– 557.780980781351
– 424.062740635737
– 127.987138247396 $\pm$ j 32.750959185172
– 67.791476256437
– 66.940000009033
– 45.689999998661
– 42.997168378168
– 26.311678722911 $\pm$ j 9.297227877312

which verifies that the closed loop system is nominally stable.

For our analysis we first took the controller coefficient vector as a parameter vector and determined the l^2 parametric stability margin around the nominal controller. This parametric stability margin is found to be:

$\rho = 1.17938672900662 \times 10^3.$

The normalized ratio of change in controller coefficients required to destabilize the closed loop is

$$\frac{\rho}{\|\mathbf{p}^0\|_2} = 1.455352715525003 \times 10^{-15}$$

which indicates that closed loop stability is very fragile with respect to controller parameter perturbations. To continue, we draw the Nyquist plot of the plant $P(s)$ with this controller $C(s)$. This is shown in Fig. 13 which shows that

Gain Margin $= [0.5745485, 2.585314]$

Phase Margin $= [0, \pm 24.0555]$ degree

Comparing this with the previous fragility with respect to coefficient perturbations reminds us that good gain and phase margins are not necessarily reliable indicators of robustness. However poor gain and/or phase margins *are* accurate indicators of fragility!

In the examples we have presented it can be seen that high order controllers indeed weaken the closed loop system even when they are designed to be robust against plant perturbations. It is worth reflecting on what this means regarding the behavior of the Nyquist plot of the plant controller pair. Indeed it suggests that as in Black's amplifier plant robustness can only be obtained at the cost of controller fragility.

9. Robust parametric synthesis: modern PID control

The demonstration of fragility of high order controllers in 1997 led to a resurgence of interest in low order controllers and in particular PID controllers. This led to the period of modern PID control, which started in 1997. These results complemented and built upon the classical results of Ziegler and Nichols (1942) and those of Åström and Hägglund (2006).

In the next subsection, we introduce PID control as a wonderful application of high gain feedback to the robust tracking and disturbance rejection problem.

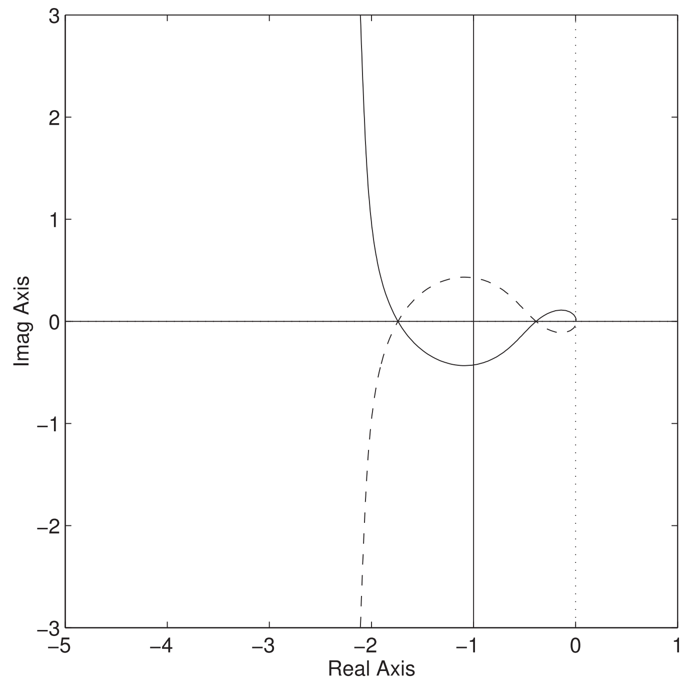

Fig. 13. Nyquist plot of $P(s)C(s)$.

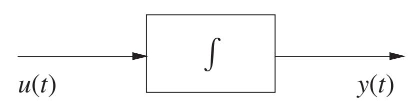

Fig. 14. An integrator.

9.1. Robustness and integral control

Integral control is a form of high gain control. It has infinite gain at DC, that is at $s = 0$. Integral control is used almost universally in the control industry to design robust servomechanisms. It works magically to remove tracking errors in the presence of disturbances even when very little is known regarding the signals and system characteristics. Integral action is most easily implemented by computer control. It turns out that hydraulic, pneumatic, electronic and mechanical integrators are also commonly used elements in control systems. In this section we explain how integral control works in general to achieve robust tracking and disturbance rejection.

Let us first consider an integrator (see Fig. 14): The input-output relationship is:

$$y(t) = K \int_0^t u(\tau) d\tau + y(0) \tag{56}$$

or

$$\frac{dy(t)}{dt} = Ku(t) \tag{57}$$

where K is the integrator gain.

Now suppose that the output $y(t)$ is a constant. It follows from (57) that

$$\frac{dy(t)}{dt} = 0 = Ku(t), \quad \text{for all } t > 0. \tag{58}$$

Eq. (58) proves the following important facts about the operation of an integrator:

(1) If the output of an integrator is constant over a segment of time, then the input must be identically zero over that same segment.

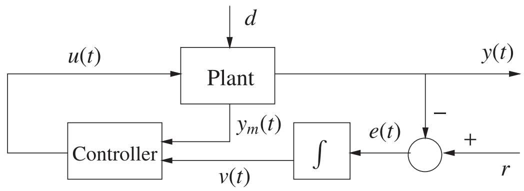

Fig. 15. Servomechanism.

(2) The output of an integrator changes as long as the input is nonzero.
The simple fact stated above suggests how an integrator can be used to solve the servomechanism problem. If a plant output $y(t)$ is to track a constant reference value r, despite the presence of unknown constant disturbances, it is enough to:
(a) attach an integrator to the plant and make the error

$$e(t) = r - y(t), \tag{59}$$

the input to the integrator.
(b) ensure that the closed loop system is asymptotically stable so that under constant reference and disturbance inputs, all signals, including the integrator output, reach constant steady state values.

This is depicted in the block diagram shown in Fig. 15.

If the feedback system shown in Fig. 15 is asymptotically stable, and the inputs r and d are constant, it follows that all signals in the closed loop will tend to constant values. In particular the integrator output $v(t)$ tends to a constant value. Therefore by the fundamental fact about the operation of an integrator established above, it follows that the integrator input tends to zero. Since we have arranged that this input is the tracking error it follows that $e(t) = r - y(t)$ tends to zero and hence $y(t)$ tracks r as $t \to \infty$.

We emphasize that the steady state tracking property established above is very robust. It holds as long as the closed loop is asymptotically stable and (1) is independent of the particular values of the constant disturbances or references, (2) is independent of the initial conditions of the plant and controller and (3) is independent of whether the plant and controller are linear or nonlinear. Thus the tracking problem is reduced to guaranteeing that stability is assured. In many practical systems stability of the closed loop system can even be ensured without detailed and exact knowledge of the plant characteristics and parameters, and this is known as robust stability.

We next discuss how several plant outputs $y_1(t), y_2(t), \ldots, y_m(t)$ can be pinned down to prescribed but arbitrary constant reference values $r_1, r_2, \ldots, r_m$ in the presence of unknown but constant disturbances $d_1, d_2, \ldots, d_q$. The previous argument can be extended to this multivariable case by attaching m integrators to the plant, and driving each integrator with its corresponding error input $e_i(t) = r_i - y_i(t),\ i = 1, \ldots, m$. This is shown in the configuration shown in Fig. 16.

Once again it follows that as long as the closed loop system is stable, all signals in the system must tend to constant values and integral action forces the $e_i(t)$, $i = 1, \ldots, m$ to tend to zero asymptotically, regardless of the actual values of the disturbances d_j, $j = 1, \ldots, q$. The existence of steady state inputs $u_1(t), u_2(t), \ldots, u_r(t)$ that make $y_i(t) = r_i,\ i = 1, \ldots, m$ for arbitrary r_i, $i = 1, \ldots, m$ requires that the plant equations relating $y_i(t)$, $i = 1, \ldots, m$ to $u_j(t)$, $j = 1, \ldots, r$ be invertible for constant inputs. In the case of linear time invariant systems this is equivalent to the requirement that the corresponding transfer matrix have rank equal to m at $s = 0$. Sometimes this is restated as the two conditions (1) $r = m$ or at least as many control inputs as outputs to be controlled and (2) $G(s)$ has no transmission zero at $s = 0$.

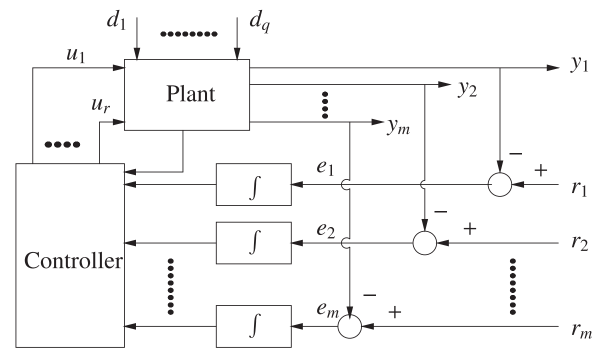

Fig. 16. Multivariable servomechanism.

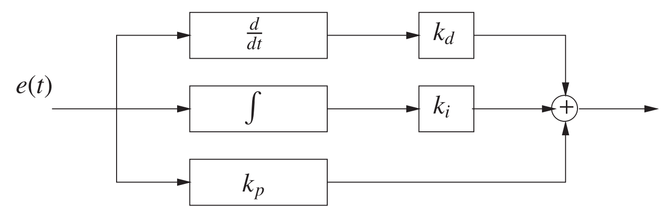

Fig. 17. PID controller.

In general, the addition of an integrator to the plant tends to make the system less stable. This is because the integrator is an inherently unstable device; for instance, its response to a step input, a bounded signal, is a ramp, an unbounded signal. Therefore the problem of stabilizing the closed loop becomes a critical issue even when the plant is stable to begin with. Since integral action and thus the attainment of zero steady state error is independent of the particular value of the integrator gain K, we can see that this gain can be used to try to stabilize the system. This single degree of freedom is sometimes insufficient for attaining stability and an acceptable transient response, and additional gains are introduced as explained in the next section. This leads naturally to the proportional integral derivative (PID) controller structure commonly used in industry (Fig. 17).

As long as the closed loop is stable it is clear that the input to the integrator will be driven to zero independent of the values of the gains. Thus the function of the gains k_p, k_i and k_d is to stabilize the closed loop system if possible, and to adjust the transient response of the system. In general the derivative can be computed or obtained if the error is varying slowly. Since the response of the derivative to high frequency inputs is much higher than its response to slowly varying signals the derivative term is usually omitted, if the error signal is corrupted by high frequency noise.

9.2. Stabilizing sets

A number of significant results on PID stabilization have been obtained in Datta et al. (2013), Silva et al. (2007), Bhattacharyya et al. (2009). These results could assist the industrial practitioner to carry out computer-aided designs of PID controllers with guaranteed stability and performance. Given the widespread use of PID controllers in various industries the impact of these results could be enormous.

Consider the general feedback system shown in Fig. 18.

Here $r(t)$ is the command signal, $y(t)$ is the output, $G(s)$ is the plant to be controlled, and $C(s)$ is the controller to be designed. The controller $C(s)$ will be assumed to be of the PID type so that

$$C(s) = k_p + \frac{k_i}{s} + k_d s \tag{60}$$

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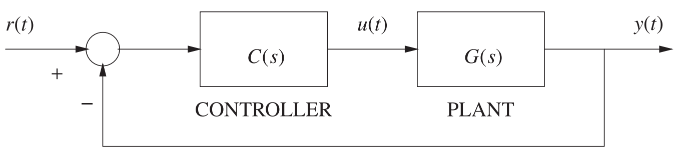


Fig. 18. Feedback control system.

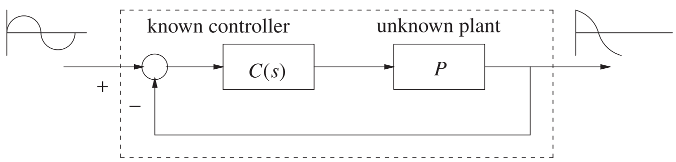


Fig. 19. Frequency response measurement on an unstable plant.

where k_p, k_i and k_d are the proportional, integral and derivative gains respectively. In many cases the pure derivative term is not allowed, particularly if the error is measured in a noisy environment. In such cases we may consider the PID controller with transfer function

$$C(s) = \frac{sk_p + k_i + k_d s^2}{s(1+sT)}, \qquad T > 0 \tag{61}$$

where T is usually fixed at a small positive value.

The plant transfer function $G(s)$ will be assumed to be rational so that

$$G(s) = \frac{N(s)}{D(s)} \tag{62}$$

where $N(s)$, $D(s)$ are polynomials in the Laplace variable s. With this plant description, the closed-loop characteristic polynomial becomes

$$\delta(s, k_p, k_i, k_d) = sD(s) + (k_i + k_d s^2)N(s) + k_p sN(s). \tag{63}$$

or with $\hat{D}(s) = D(s)(1+sT)$,

$$\delta(s, k_p, k_i, k_d) = s\hat{D}(s) + (k_i + k_d s^2)N(s) + k_p sN(s). \tag{64}$$

depending on the particular type of $C(s)$ being discussed. The problem of stabilization using a PID controller is to determine the values of k_p, k_i and k_d for which the closed-loop characteristic polynomial is Hurwitz, that is, has all its roots in the open left half plane. Since plants with a zero at the origin ($N(0) = 0$) cannot be stabilized by PID controllers we exclude such plants at the outset. Finding the complete set of stabilizing parameters is an important first step in searching for subsets attaining various design objectives. We will now develop an algorithm for computationally characterizing this set for a given plant with a rational transfer function (Fig. 19).

The set of controllers of a given structure that stabilizes the closed loop is of fundamental importance since every design must belong to this set and any performance specifications that are imposed must be achieved over this set.

Write

$$\mathbf{k} := [k_p, k_i, k_d] \tag{65}$$

and let

$$\mathcal{S}^o := \{\mathbf{k} : \delta(s, \mathbf{k}) \text{ is Hurwitz}\} \tag{66}$$

denote the set of PID controllers that stabilize the closed-loop for the given plant characterized by the transfer function $(N(s), D(s))$. It is emphasized that due to the presence of integral action on the error any controller in $\mathcal{S}^o$ already provides asymptotic tracking and disturbance rejection for step inputs. In general, additional design specifications on stability margins and transient response are also required and subsets representing these must be sought within $\mathcal{S}^o$.

The three dimensional set $\mathcal{S}^o$ is simply described by (66) but not necessarily simple to calculate. For example, a naive application of the Routh–Hurwitz criterion to $\delta(s, \mathbf{k})$ will result in a description of $\mathcal{S}^o$ in terms of highly nonlinear and intractable inequalities.

In the following, we describe a computationally efficient approach, developed in Datta et al. (2013), Silva et al. (2007), Bhattacharyya et al. (2009), to determine PID stabilizing sets using frequency response data and the concepts of "signature".

9.3. Phase, signature, poles, zeros, and Bode plots

Let P denote a LTI plant and $P(s)$ its rational transfer function with z^+, p^+ (z^-, p^-) denoting the numbers of RHP (LHP) zeros and poles, $n(m)$ the denominator (numerator) degrees. Let the relative degree be denoted r_P:

$$r_P := n - m$$

Define the signature of P as:

$$\sigma(P) := \left(z^- - z^+\right) - \left(p^- - p^+\right). \tag{67}$$

Lemma 9.1. *Keel and Bhattacharyya (2008)*

$$r_P = -\frac{1}{20} \cdot \left.\frac{dP_{\text{db}}(\omega)}{d(\log_{10}\omega)}\right|_{\omega\to\infty} \tag{68}$$

$$\sigma(P) = \frac{2}{\pi}\Delta_0^\infty \angle P(j\omega) \tag{69}$$

where

$$P_{\text{db}}(\omega) := 20\log_{10}|P(j\omega)|.$$

Assuming that $P(s)$ has no $j\omega$ axis poles and zeros, we can also write

$$\sigma(P) = -(n-m) - 2\left(z^+ - p^+\right) \tag{70}$$

or

$$\sigma(P) = -r_P - 2\left(z^+ - p^+\right). \tag{71}$$

Therefore, $z^+ - p^+$ can be determined from the Bode plot of P. In particular if $P(s)$ is stable the Bode plot can often be obtained experimentally by measuring the frequency response of the system. Then the above relations with $p^+ = 0$ determine z^+ from the Bode plot data.

Now suppose that P is an *unstable* LTI plant with a rational transfer function *unknown* to us and assume that P does not have imaginary axis poles and zeros. We assume however that a known feedback controller $C(s)$ stabilizes P and the closed-loop frequency response can be *measured* and is denoted by $G(j\omega)$ for $\omega \in [0, \infty)$: Then

$$P(j\omega) = \frac{G(j\omega)}{C(j\omega)(1 - G(j\omega))} \tag{72}$$

is the *computed* frequency response of the unstable plant. The next result shows that knowledge of $C(s)$ and $G(j\omega)$ is sufficient to determine the numbers z^+ and p^+, that is the numbers of RHP zeros and poles of the plant. Let z_c^+ denote the number of RHP zeros of $C(s)$.

Theorem 9.2. *Keel and Bhattacharyya (2008)*

$$z^+ = \frac{1}{2}\left[-r_P - r_C - 2z_c^+ - \sigma(G)\right] \tag{73}$$

$$p^+ = \frac{1}{2}[\sigma(P) - \sigma(G) - r_C] - 2z_c^+. \tag{74}$$

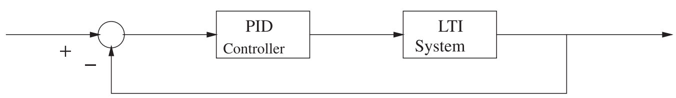


Fig. 20. A unity feedback system with a PID controller.

Sketch of proof

We have

$$G(s) = \frac{P(s)C(s)}{1+P(s)C(s)}$$

and, since $G(s)$ is stable,

$$\begin{aligned}\sigma(G) &= \left(z^- + z_c^-\right) - \left(z^+ + z_c^+\right) - (n+n_c) \\ &= -r_P - r_C - 2z_c^+ - 2z^+\end{aligned}$$

which implies (73). From (67) applied to $P(s)$, we have

$$p^+ = z^+ + \frac{\sigma(P)}{2} + \frac{r_P}{2}. \tag{75}$$

Substituting (73) in (75), we have (74).

Remark 9.3. In the above theorem, we assume that $C(s)$, a stabilizing controller is known, and the corresponding closed-loop frequency response $G(j\omega)$ for $\omega \in [0, \infty)$ can be *measured*. Thus, $P(j\omega)$ can be computed from (72). Now r_P and $\sigma(G)$ can be found by applying the results of Lemma 9.1 to $P(j\omega)$ and $G(j\omega)$, respectively. r_C and z_c^+ are known as $C(s)$ is known. Therefore, z^+ and p^+ can be found.

Remark 9.4. In the above analysis we have assumed for simplicity that the plant is devoid of imaginary axis poles and zeros. When such poles and zeros are present their numbers may be known from physical considerations or their numbers and locations may be ascertained from the experimentally determined or computed Bode plot. Once identified we can lump these poles and zeros with the controller and proceed with the design procedure. At an imaginary axis zero (pole) of multiplicity k, away from the origin, the magnitude plot is zero (infinity) and the phase plot undergoes an instantaneous change of phase $k\pi$. If such poles or zeros occur at the origin there is a corresponding phase shift of $\frac{k\pi}{2}$ at zero frequency. It is straightforward to establish that, in this case, the relations (73) and (74) need to be modified to the following:

$$z^+ = \frac{1}{2}\left[-r_P - r_C - 2z_c^+ - z_c^i - \sigma(G)\right] \tag{76}$$

$$p^+ = \frac{1}{2}[\sigma(P) - \sigma(G) - r_C] - 2z_c^+ - z_c^i + z^i - p^i. \tag{77}$$

where z^i, z_c^i, p^i, p_c^i denote the numbers of imaginary axis zeros and poles of plant and controller.

10. PID synthesis for delay free continuous-time systems

In this section, we consider the synthesis and design of PID controllers for a continuous-time LTI plant, with underlying transfer function $P(s)$ with $n(m)$ poles (zeros). (see Fig. 20). We assume that the *only* information available to the designer is:

1. Knowledge of the frequency response magnitude and phase, equivalently, $P(j\omega)$, $\omega \in [0, \infty)$ if the plant is stable.
2. Knowledge of a known stabilizing controller and the corresponding closed-loop frequency response $G(j\omega)$.

Such assumptions are reasonable for most systems. We also make the technical assumption that the plant has no $j\omega$ poles or zeros so that the magnitude, its inverse and phase are well-defined for all $\omega \geq 0$. As we have seen from the discussion in the last section, the numbers and locations of RHP poles and zeros can be found from the above data for any LTI plant and either "divided out" or lumped with the controller.

Write

$$P(j\omega) = |P(j\omega)|e^{j\phi(\omega)} = P_r(\omega) + jP_i(\omega) \tag{78}$$

where $|P(j\omega)|$ denotes the *magnitude* and $\phi(\omega)$ the *phase* of the plant, at the frequency ω.

Let the PID controller be of the form

$$C(s) = \frac{K_i + K_p s + K_d s^2}{s(1+sT)}, \quad \text{for } T > 0 \tag{79}$$

where T is assumed to be fixed and small. We now present results for developing our procedure for determining the stabilizing set.

Lemma 10.1. Keel and Bhattacharyya (2008) *Let*

$$F(s) := s(1+sT) + \left(K_i + K_p s + K_d s^2\right)P(s).$$

and

$$\bar{F}(s) := F(s)P(-s).$$

Then closed-loop stability is equivalent to

$$\sigma\left(\bar{F}(s)\right) = n - m + 2z^+ + 2. \tag{80}$$

Write

$$\begin{aligned}\bar{F}(j\omega) &= j\omega(1+j\omega T)P(-j\omega) \\ &\quad + \left(K_i + j\omega K_p - \omega^2 K_d\right)P(j\omega)P(-j\omega) \\ &= \underbrace{\left(K_i - K_d\omega^2\right)|P(j\omega)|^2 - \omega^2 T P_r(\omega) + \omega P_i(\omega)}_{\bar{F}_r(\omega, K_i, K_d)} \\ &\quad + j\omega\underbrace{\left(K_p|P(j\omega)|^2 + P_r(\omega) + \omega T P_i(\omega)\right)}_{\bar{F}_i(\omega, K_p)} \\ &= \bar{F}_r(\omega, K_i, K_d) + j\omega\bar{F}_i(\omega, K_p).\end{aligned}$$

Theorem 10.2. Keel and Bhattacharyya (2008) *Let $\omega_1 < \omega_2 < \cdots < \omega_{l-1}$ denote the distinct frequencies of odd multiplicities which are solutions of*

$$\bar{F}_i(\omega, K_p^*) = 0, \tag{81}$$

or

$$\begin{aligned}K_p^* &= -\frac{P_r(\omega) + \omega T P_i(\omega)}{|P(j\omega)|^2} \\ &= -\frac{\cos\phi(\omega) + \omega T \sin\phi(\omega)}{|P(j\omega)|} \\ &= : g(\omega)\end{aligned} \tag{82}$$

for fixed $K_p = K_p^*$. Let $\omega_0 = 0$ and $\omega_l = \infty$, and $j := \operatorname{sgn}\left[\bar{F}_i(\infty^-, K_p^*)\right]$. Determine strings of integers

$$I = [i_0, i_1, i_2, \ldots, i_l],$$

with $i_t \in \{+1, -1\}$ such that

For $n - m$ even:

$$\begin{aligned}&\left[i_0 - 2i_1 + 2i_2 + \cdots + (-1)^{l-1}2i_{l-1} + (-1)^l i_l\right](-1)^{l-1}j \\ &\quad = n - m + 2z^+ + 2.\end{aligned} \tag{83}$$

For $n - m$ odd

$$\begin{aligned}&\left[i_0 - 2i_1 + 2i_2 + \cdots + (-1)^{l-1}2i_{l-1}\right](-1)^{l-1}j \\ &\quad = n - m + 2z^+ + 2.\end{aligned} \tag{84}$$

Then for the fixed $K_p = K_p^*$, the (K_i, K_d) values corresponding to closed-loop stability are given by

$$\bar{F}_r(\omega_t, K_i, K_d)i_t > 0, \tag{85}$$

where the i_t's are taken from strings satisfying (83) and (84), and ω_t's are taken from the solutions of (81).

The following result clarifies what range the parameters K_p must be swept over.

Theorem 10.3. *Keel and Bhattacharyya (2008) For the given function $g(\omega)$ in (82) determined completely by the plant data $P(\jmath\omega)$ and T:*

(1) *A necessary condition for PID stabilization is that there exists K_p such that the function*

$$K_p = g(\omega) \tag{86}$$

has at least k distinct roots of odd multiplicities where

$$k \geq \frac{n-m+2z^+ + 2}{2} - 1 \quad \text{if } n-m \text{ even}$$

$$k \geq \frac{n-m+2z^+ + 3}{2} - 1 \quad \text{if } n-m \text{ odd}.$$

(2) *There exists unique range ω, $\underline{\omega} = (\omega_{\min}, \omega_{\max})$ over which the condition (1) occurs and this determines the range of ω to be swept.*

(3) *Every K_p belonging to the stabilizing set of PID parameters lies in the range*

$$K_p \in \left(K_p^{\min}, K_p^{\max}\right)$$

where

$$K_p^{\min} := \min_{\omega \in \underline{\omega}} g(\omega), \qquad K_p^{\max} := \max_{\omega \in \underline{\omega}} g(\omega)$$

where $\underline{\omega} = (\omega_{\min}, \omega_{\max})$.

Remark 10.4. The function $g(\omega)$ is well defined due to the assumption that the plant either has no $\jmath\omega$ zeros or these have been lumped with the controller. The frequency $\omega_{\max}$ can be selected as any frequency after which $g(\omega)$ continues to grow monotonically. This determines the range of frequencies over which the $P(\jmath\omega)$ data of the plant should be known accurately. Note that the ranges $(\omega_{\min}, \omega_{\max})$ and $(K_p^{\min}, K_p^{\max})$ can consist of multiple segments.

The computation of the stabilizing set implied by the above results is summarized as the following procedure.

Computation of PID stabilizing sets from Nyquist/Bode data

The complete set of stabilizing PID gains can be computed by the following procedure:

For stable systems : The available data is the frequency response of the plant $P(\jmath\omega)$.

0.1 Determine the relative degree of the plant $r_P = n - m$ from the high frequency slope of the Bode magnitude plot of $P(\jmath\omega)$.

0.2 Let $\Delta_0^\infty[\phi(\omega)]$ denote the net change of phase of $P(\jmath\omega)$ for $\omega \in [0, \infty)$. Determine z^+ from

$$\Delta_0^\infty[\phi(\omega)] = -\left[(n-m) + 2z^+\right]\frac{\pi}{2}$$

which follows from (70) with $p^+ = 0$.

For unstable systems : The available data are a stabilizing controller transfer function $C(s)$ and the frequency response of the corresponding stable closed-loop system $G(\jmath\omega)$.

0.1 Compute the frequency response $P(\jmath\omega)$ from (72).
0.2 Determine relative degree of the plant r_P from the high frequency slope of the Bode magnitude plot of $P(\jmath\omega)$.
0.3 Determine z_c^+ and r_C from $C(s)$.
0.4 Compute $\sigma(G)$ from (69) applied to $G(\jmath\omega)$.
0.5 Compute z^+ using (73) in Theorem 9.2.
0.6 Compute $g(\omega)$ using (82) and the data.

1. Fix $K_p = K_p^*$, solve (82) and let $\omega_1 < \omega_2 < \cdots < \omega_{l-1}$ denote the distinct frequencies of odd multiplicities which are solutions of (82).

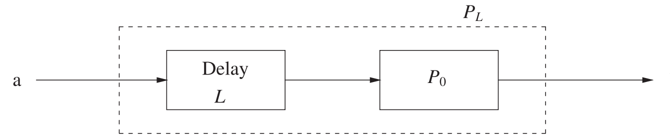


Fig. 21. A plant with cascaded delay.

2. Set $\omega_0 = 0$, $\omega_l = \infty$ and $j = \text{sgn}\bar{F}_i(-\infty^-, K_p^*)$. Determine all strings of integers $i_t \in \{+1, -1\}$ such that: For $n-m$ even:

$$\left[i_0 - 2i_1 + 2i_2 + \cdots + (-1)^{i-1}2i_{l-1} + (-1)^l i_l\right](-1)^{l-1} j = n - m + 2z^+ + 2. \tag{87}$$

For $n-m$ odd:

$$\left[i_0 - 2i_1 + 2i_2 + \cdots + (-1)^{i-1}2i_{l-1}\right](-1)^{l-1} j = n - m + 2z^+ + 2. \tag{88}$$

3. For the fixed $K_p = K_p^*$ chosen in Step 1, solve for the stabilizing (K_i, K_d) values from

$$\left[K_i - K_\text{d}\omega_t^2 + \frac{\omega_t \sin\phi(\omega_t) - \omega_t^2 T \cos\phi(\omega_t)}{|P(\jmath\omega_t)|}\right] i_t > 0, \tag{89}$$

for $t = 0, 1, \ldots, l$.

4. Repeat the previous three steps by updating K_p over prescribed ranges. The ranges over which K_p must be swept is determined from the requirements that (87) or (88) is satisfied for at least one string of integers as in Theorem 10.3.

We emphasize that all computations are based on the data $P(\jmath\omega)$ and knowledge of the transfer function $P(s)$ is not required. For well-posedness of the loop, it is necessary that

$$K_\text{d} \neq -\frac{T}{P(\infty)}.$$

For strictly proper plants, $P(\infty) = 0$ and this constraint vanishes.

11. PID controller synthesis for systems with delay

In this section, we show how the previous results can be extended to systems with delay. Consider the finite dimensional LTI plant P_L with a cascaded delay in Fig. 21. Here P_0 represents an LTI delay free system with a proper transfer function. The transfer functions of P_0 and P_L are denoted $P_0(s)$ and $P_L(s)$, respectively. We assume that frequency response measurements can be made at terminals "a" and "b," that is on the delay system P_L. Thus, the data we have is:

$$P_L(\jmath\omega) = e^{-\jmath\omega L}P_0(\jmath\omega) = m_L(\omega)e^{\jmath\phi_L(\omega)}, \quad 0 \leq \omega < \infty.$$

Write

$$P_0(\jmath\omega) = m_0(\omega)e^{\jmath\phi_0(\omega)}.$$

Therefore,

$$m_0(\omega) = m_L(\omega), \tag{90}$$

and

$$\phi_L(\omega) = \phi_0(\omega) - \omega L, \quad 0 \leq \omega < \infty. \tag{91}$$

It is clear that $m_0(\omega)$ and $\phi_0(\omega)$ can be determined from the data $m_L(\omega)$ and $\phi_L(\omega)$ on the system with embedded delay L when L is known. If L is unknown it can be determined from the high frequency slope of the phase.

Now let $C(s, \mathbf{k})$ denote the PID controller

$$C(s, \mathbf{k}) = \frac{K_i + K_p s + K_d s^2}{s(1+sT)}, \qquad \mathbf{k} = [K_i,\ K_p,\ K_\text{d}].$$

Let $\mathcal{S}_0$ denote the set of stabilizing PID controllers for the delay free plant

$S_0 = \{\mathbf{k} \; : \; C(s, \mathbf{k}) \text{ stabilizes } P_0\}.$

We have seen how $\mathcal{S}_0$ can be found in the previous section when $P_0(\jmath\omega)$ is known. It follows from (91) that $\mathcal{S}_0$ can be determined from $P_L(\jmath\omega)$ the data for the embedded delay system. We denote by $\mathcal{S}_L$ the set of PID controllers stabilizing the plant P_0 with cascaded delay ranging from 0 to L seconds.

Following the results in Chapter 5, introduce the sets

$$\mathcal{S}_\infty = \{\mathbf{k} \; : \; |C(s, \mathbf{k})P_0(s)|_{s=\infty} \geq 1\},$$
$$\mathcal{S}_\mathcal{B} = \left\{\mathbf{k} \; : \; C(\jmath\omega, \mathbf{k}) = \frac{-e^{\jmath\omega \underline{L}}}{P_0(\jmath\omega)}, \quad \begin{matrix} \omega \in [0, \infty), \\ 0 \leq \underline{L} \leq L \end{matrix}\right\}.$$

The following theorem is obtained from the results of Chapter 5.

Theorem 11.1. *Keel and Bhattacharyya (2008) The set $\mathcal{S}_L$ can be found from*

$$\mathcal{S}_L = \mathcal{S}_0 \backslash (\mathcal{S}_\infty \cup \mathcal{S}_\mathcal{B}). \tag{92}$$

The set $\mathcal{S}_\infty$ consists of those PID gains for which the Nyquist plot approaches points outside the unit circle as $s \to \infty$. The set $\mathcal{S}_\mathcal{B}$ consists of those PID gains for which an imaginary axis characteristic root occurs for delays less than L. The relationship (92) states that the exclusion of $\mathcal{S}_\infty$ and $\mathcal{S}_\mathcal{B}$ from $\mathcal{S}_0$ determines the stabilizing set $\mathcal{S}_L$ for the system with cascaded delay up to L seconds.

The computation of $\mathcal{S}_0$ has been described in the previous section. The set $\mathcal{S}_\infty$ is easy to calculate. In fact,

$$\mathcal{S}_\infty = \left\{\mathbf{k} \; : \; |K_\mathrm{d}| \geq \frac{T}{|P_0(\infty)|}\right\}.$$

To determine $\mathcal{S}_\mathcal{B}$, let

$$\theta(\omega, \mathbf{k}) := \angle\left(\frac{K_i - K_\mathrm{d}\omega^2 + \jmath\omega K_p}{\jmath\omega(1 + \jmath\omega T)}\right).$$

Then the conditions defining $\mathcal{S}_\mathcal{B}$ can be written as the magnitude and phase conditions

$$K_i - K_\mathrm{d}\omega^2 = \pm\sqrt{\frac{\omega^2(1+T^2)}{m_0^2(\omega)} - \omega^2 K_p^2}, \tag{93}$$

$$\theta(\omega, \mathbf{k}) \geq \pi - \phi_0(\omega) - \omega L, \quad \omega \in [0, \infty), \tag{94}$$

so that

$\mathcal{S}_\mathcal{B} = \{\mathbf{k} \; : \; \mathbf{k} \text{ satisfies (93) and (94) for some } \omega \in [0, \infty)\}.$

Note that (93) represents a straight line in (K_i, K_d) space for each fixed K_p and ω. The calculation of the set $\mathcal{S}_\mathcal{B}$ is tedious but straightforward if one sweeps over the frequency variable.

12. Computer-aided design (Bhattacharyya et al., 2009)

In this section we show some possibilities for computer-aided design using the above theory. The algorithm for the design of a PID Controller from the frequency response data of the system has been programmed in LabVIEW due to its user-friendly graphical environment. The Virtual Instrument (VI) has a front panel that is displayed to the user and a block diagram, where the computations are performed. The inputs to the LabVIEW program are the frequency response data and the number of RHP poles of the system. Given these inputs, the entire range of K_p that can stabilize the system is displayed, and as the user scrolls through the stabilizing range of K_p, entire stabilizing ranges of K_i and K_d are displayed.

The file containing the frequency response data from a stable system (number of right-half plane poles equals zero) is fed into the program through the file path box located at the top left-hand side of the VI as shown in Fig. 22.

T is set to a very small number. The equation for $g(\omega)$ versus frequency generated from the inputs, shown in Fig. 22 is used to compute the ω values associated with a particular value of K_p. Using the values of ω's obtained above, linear equations are solved by the program to compute stabilizing sets of K_i and K_d for a fixed K_p. As the selected K_p is changed, the K_i, K_d region changes dynamically to show the new stabilizing ranges of K_i and K_d for the selected K_p. The K_i, K_d graph shown in Fig. 23. gives the stabilizing range for $K_p = 1$. Each point in the stable range of K_i and K_d for a specific K_p has performance indices, as shown in Fig. 24. The performance indices dynamically change as the cursor moves over the stable range of K_i and K_d, for a particular K_p.

The subsets within these stabilizing ranges that achieve or exceed the desired minimum performance indices can be computed by the program. The front panel of the VI that performs this operation is shown in Fig. 25 and the final result is shown in Fig. 26. The set of points generated satisfies multiple performance indices simultaneously. The performance indices used in the program are: Gain Margin$\geq$ 3db, Phase Margin$\geq$ 45^o, Overshoot$\leq$ 30%.

Similar results have been obtained for discrete-time systems, see Keel and Bhattacharyya (2006), Bhattacharyya et al. (2009).

13. Achievable performance: gain and phase margin specifications

13.1. Continuous-time controllers: constant gain and phase Loci

For continuous-time controllers, it is possible to parametrize the controller parameters in a geometric form. For the cases of PI and PID controllers, the constant gain and phase loci result in ellipses and straight lines.

13.2. PI controllers Diaz-Rodriguez and Bhattacharyya (2016)

Let $P(s)$ and $C(s)$ denote the plant and controller transfer functions. The frequency response of the plant and controller are $P(j\omega)$, $C(j\omega)$ respectively where $\omega \in [0, \infty]$. For a PI controller

$$C(s) = \frac{K_P s + K_I}{s}, \tag{95}$$

where K_P and K_I are design parameters. Then with $s = j\omega$, we have

$$|C(j\omega)|^2 = K_P^2 + \frac{K_I^2}{\omega^2} =: M^2 \tag{96}$$

$$\angle C(j\omega) = \tan^{-1}\frac{K_I}{\omega K_P} := \Phi \tag{97}$$

Eqs. (96) and (97) can be written as

$$\frac{(K_P)^2}{a^2} + \frac{(K_I)^2}{b^2} = 1, \qquad K_I = cK_P, \tag{98}$$

where $a^2 = M^2$, $b^2 = M^2\omega^2$, $c = \omega\tan\Phi$. We have ellipses and straight lines in (K_P, K_I) space. The major and minor axes of the ellipse are given by a and b. The slope of the line is represented by c. Suppose ω_g is the prescribed closed-loop gain crossover frequency. Then $M_g := \frac{1}{|P(j\omega_g)|}$. If ϕ_g^* is the desired phase margin in radians, $\Phi_g := \pi + \phi_g^* - \angle P(j\omega_g)$. From (98) we obtain the ellipse and straight line corresponding to $M = M_g$ and $\Phi = \Phi_g$, giving the design point (K_P^*, K_I^*). If these intersection points lie in the stabilizing set S, the design is feasible, otherwise the specifications have to be altered.

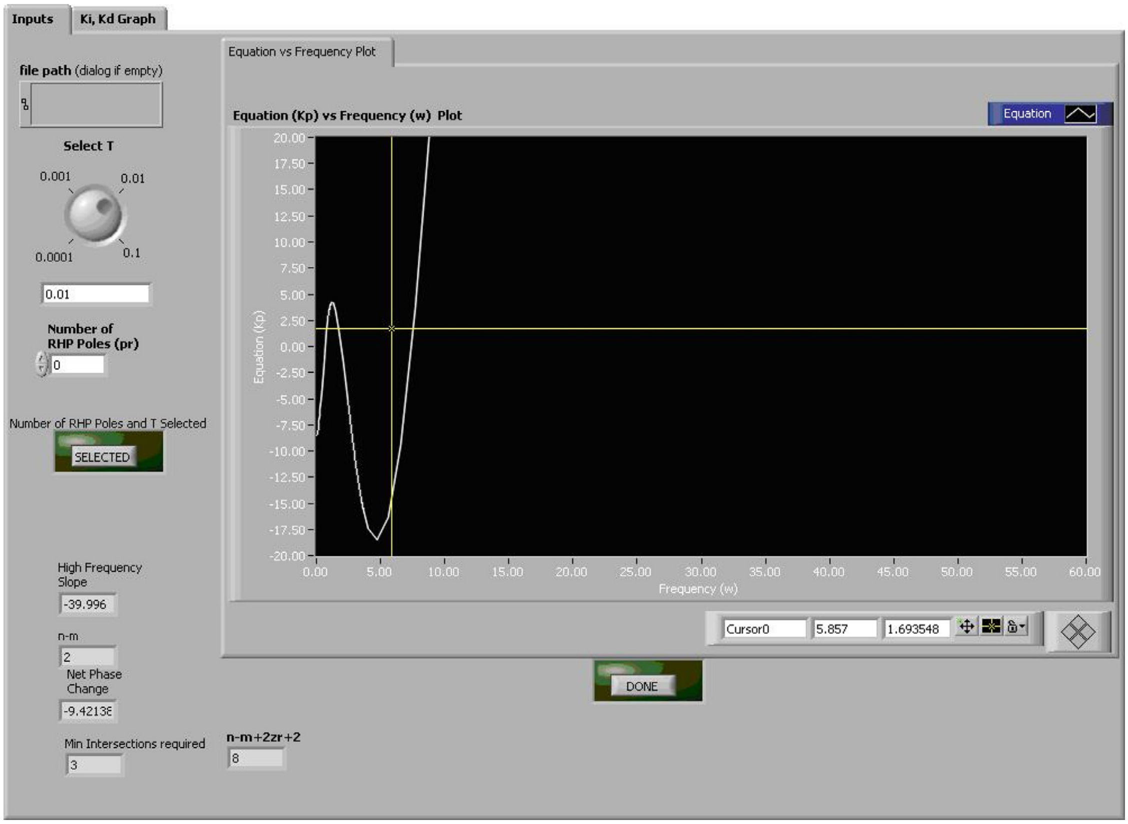


Fig. 22. Front panel of VI page 1 showing stabilizing sets of K_p, K_i, and K_d.

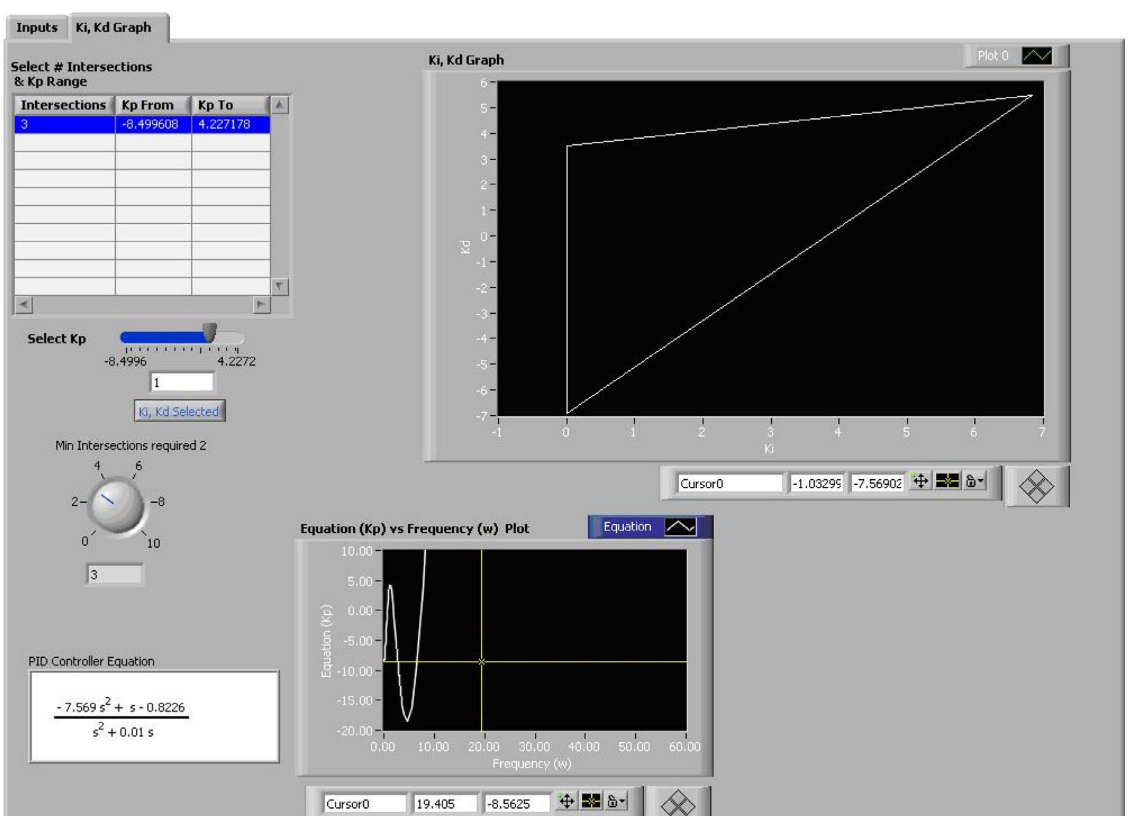


Fig. 23. Front panel of the VI page 2 showing stabilizing sets of K_p, K_i, and K_d.

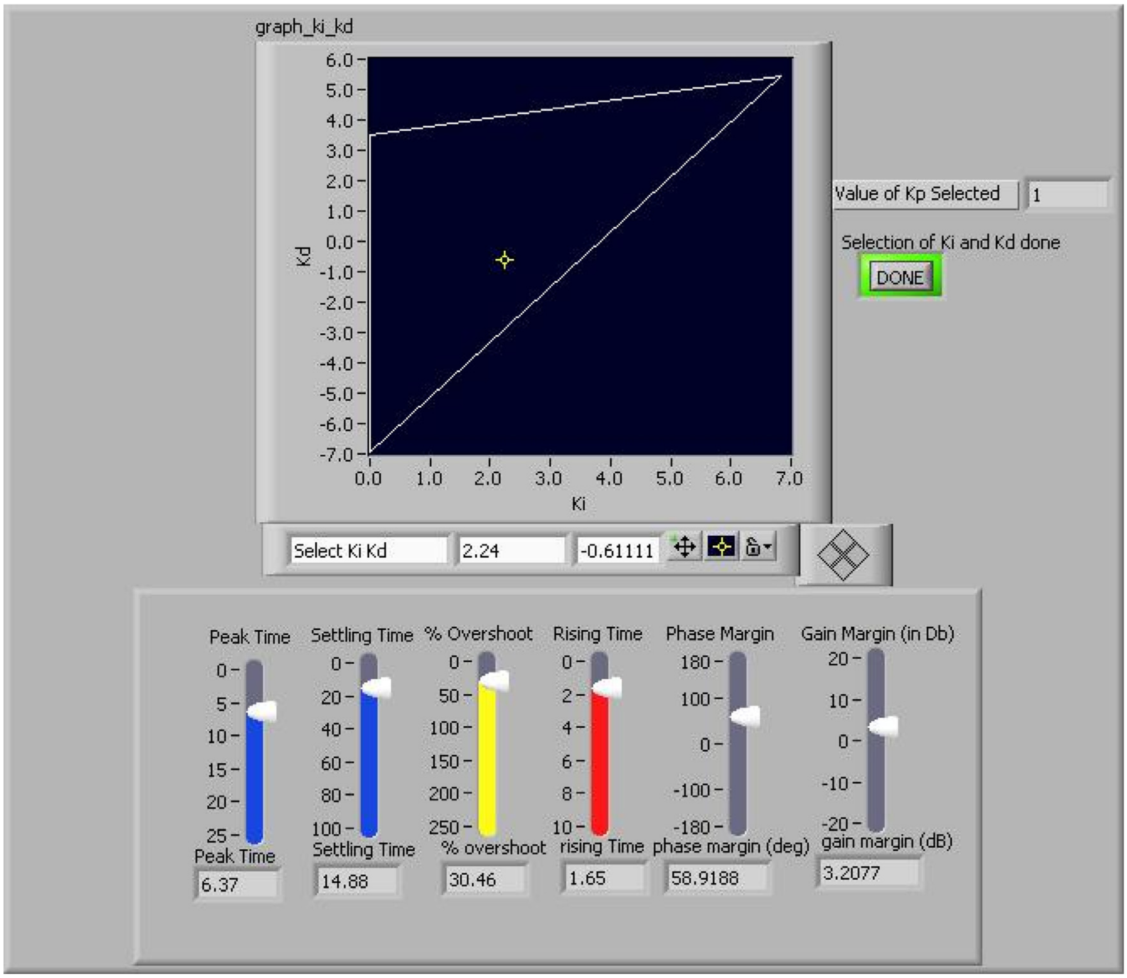


Fig. 24. Front panel of VI showing performance indices for specific K_p, K_i, and K_d.

PID controllers (Diaz-Rodriguez, Han, Lee, & Bhattacharyya, 2017)

Let $P(s)$ and $C(s)$ denote the plant and controller transfer functions. The frequency response of the plant and controller are $P(j\omega)$, $C(j\omega)$ respectively where $\omega \in [0, \infty]$. For a PID controller

$$C(s) = \frac{K_D s^2 + K_P s + K_I}{s}, \tag{99}$$

where K_P, K_I, and K_D are design parameters. Then, for $s = j\omega$

$$|C(j\omega)|^2 = K_P^2 + \left(K_D\omega - \frac{K_I}{\omega}\right)^2 := M^2 \tag{100}$$

$$\angle C(j\omega) = \tan^{-1}\left(K_D\omega - \frac{K_I}{\omega}\right) := \Phi \tag{101}$$

from (100) and (101) we have that

$$K_P^2 = M^2 - \left(K_D\omega - \frac{K_I}{\omega}\right)^2 = \frac{\left(K_D\omega - \frac{K_I}{\omega}\right)^2}{\tan^2\Phi}. \tag{102}$$

From (102) we can have the following expressions

$$K_I = K_D\omega^2 \pm \omega\sqrt{\frac{M^2\tan^2\Phi}{1+\tan^2\Phi}} \tag{103}$$

$$K_P = \pm\sqrt{\frac{M^2}{1+\tan^2\Phi}} \tag{104}$$

Suppose ω_g is the prescribed closed-loop gain crossover frequency. Then $M_g = \frac{1}{|P(j\omega_g)|}$. If ϕ_g^* is the desired phase margin in radians, $\Phi_g = \pi + \phi_g^* - \angle P(j\omega_g)$. Thus, for Eq. (103) is a straight line in (K_I, K_D) plane for a constant value of K_P represented in Eq. (104).

14. Discrete-time controllers: constant gain and phase loci

For discrete-time controllers, it is possible to parametrize the controller parameters in a geometric form. For the cases of PI and PID digital controllers, the constant gain and phase loci result in ellipses and straight lines.

14.1. PI controllers (Diaz-Rodriguez & Bhattacharyya, 2015)

Let $P(z)$ and $C(z)$ denote the plant and controller transfer functions. The frequency response of the plant and controller are $P(e^{j\omega T})$, $C(e^{j\omega T})$, respectively, where T is the sampling period and $\omega \in \left[0, \frac{2\pi}{T}\right]$. For a PI controller

$$C(z) = \frac{K_0 + K_1 z}{z-1}, \tag{105}$$

where K_0, K_1 are design parameters. Then, with $z = e^{j\omega T}$ and letting $\omega T =: \theta$

$$C(e^{j\theta}) = \frac{(K_1+K_0)\cos\frac{\theta}{2} + j(K_1-K_0)\sin\frac{\theta}{2}}{2j\sin\frac{\theta}{2}}. \tag{106}$$

Let $L_0 := K_1 + K_0$ and $L_1 := K_1 - K_0$. From this and (106) we have

$$|C(e^{j\theta})|^2 = \frac{L_0^2}{4\tan^2\frac{\theta}{2}} + \frac{L_1^2}{4} =: M^2 \tag{107}$$

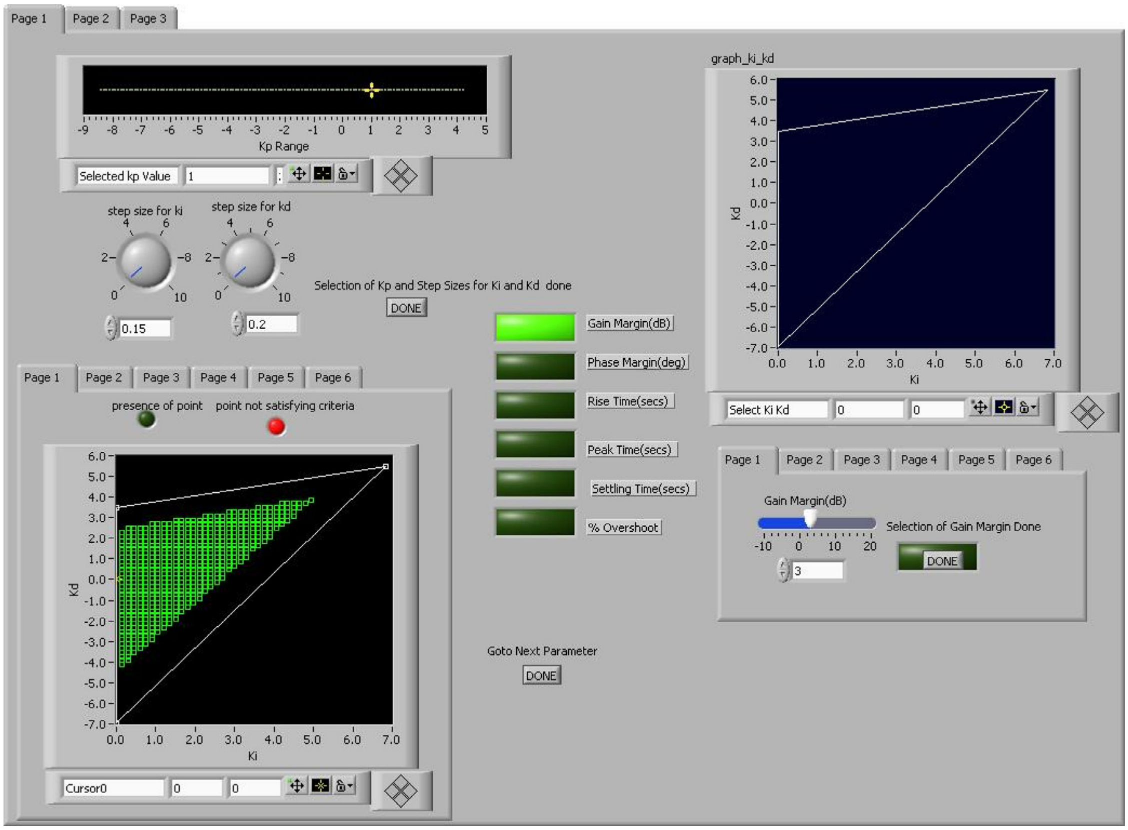


Fig. 25. Front panel of VI that satisfies multiple performance index specifications.

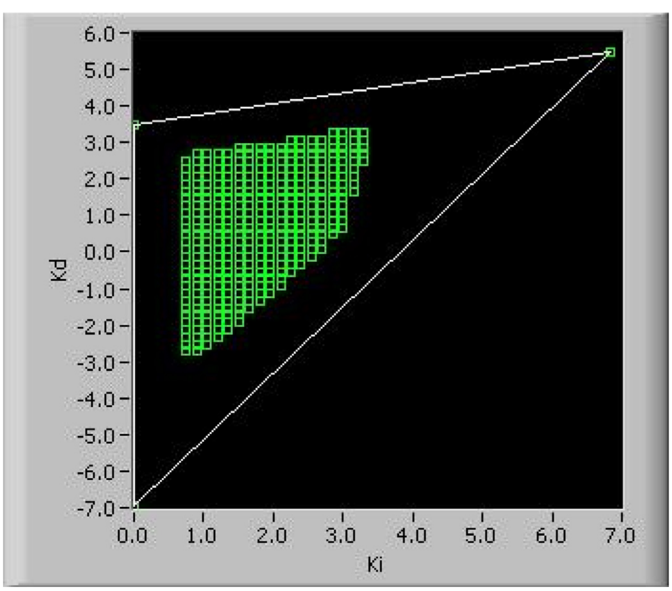


Fig. 26. LabVIEW generated result for multiple performance indices.

$$\angle C(e^{j\theta}) = \tan^{-1}\frac{-L_0}{L_1\tan\frac{\theta}{2}} =: \Phi \tag{108}$$

Eqs. (107) and (108) can be written as

$$\frac{L_0^2}{a^2} + \frac{L_1^2}{b^2} = 1, \qquad L_1 = cL_0 \tag{109}$$

where

$$a^2 = 4M^2\tan^2\frac{\theta}{2}, \quad b^2 = 4M^2 \quad c = -\frac{1}{\tan\Phi\tan\frac{\theta}{2}} \tag{110}$$

We have ellipses and straight lines in L_0, L_1 space. The major and minor axes of the ellipse are given by a and b, c represents the slope of the straight line. The mapping from K_0, K_1 to L_0, L_1 and vice versa is given by

$$K_0 = \frac{L_0 - L_1}{2}, \qquad K_1 = \frac{L_0 + L_1}{2}. \tag{111}$$

Suppose ω_g is the prescribed closed-loop gain crossover frequency. Then $M_g := \frac{1}{|P(e^{j\theta_g})|}$ and if ϕ_g^* is the desired phase margin in radians, $\Phi_g := \pi + \phi_g^* - \angle P(e^{j\theta_g})$. From (109) we obtain the ellipse and straight lines corresponding to $M = M_g$ and $\Phi = \Phi_g$, giving the design point (K_P^*, K_I^*). If these intersection points lie in the stabilizing set S, the design is feasible, otherwise the specifications have to be altered.

14.2. PID controllers (Diaz-Rodriguez et al., 2015)

Let $P(z)$ and $C(z)$ denote the plant and controller transfer functions. The frequency response of the plant and controller are $P(e^{j\omega T})$, $C(e^{j\omega T})$ respectively where T is the sampling period and $\omega \in \left[0, \frac{2\pi}{T}\right]$. For a PID controller

$$C(z) = \frac{K_0 + K_1 z + K_2 z^2}{z(z-1)}, \tag{112}$$

where K_0, K_1, and K_2 are design parameters. Then, with $z = e^{j\omega T}$ and letting $\omega T =: \theta$

$$C(e^{j\theta}) = \frac{(K_2 + K_0)\cos\theta + K_1 + j(K_2 - K_0)\sin\theta}{(\cos\theta - 1 + j\sin\theta)}. \tag{113}$$

Let $L_0 := K_2 + K_0,\ L_1 := K_2 - K_0$. From this and (113) we have

$$|C(e^{j\theta})|^2 = \frac{\left(L_0 + \frac{K_1}{\cos\theta}\right)^2}{\left(\frac{\sqrt{\mu}}{\cos\theta}\right)^2} + \frac{L_1^2}{\left(\frac{\sqrt{\mu}}{\sin\theta}\right)^2} =: M^2 \tag{114}$$

$$\angle C(e^{j\theta}) = \tan^{-1}\left(\frac{\sin\theta(L_1(\cos\theta-1) - (L_0\cos\theta + K_1))}{(L_0\cos\theta + K_1)(\cos\theta - 1) + L_1\sin^2\theta}\right) =: \Phi. \tag{115}$$

where $\mu = (\cos\theta - 1)^2 + (\sin\theta)^2$. Eqs. (114) and (115) can be written as

$$\frac{(L_0 + a)^2}{b^2} + \frac{L_1^2}{c^2} = 1, \qquad L_1 = dL_0 + e \tag{116}$$

where

$$a = \frac{K_1}{\cos\theta}, \quad b^2 = \frac{\mu M^2}{\cos^2\theta}, \quad c^2 = \frac{\mu M^2}{\sin^2\theta} \tag{117}$$

$$d = \frac{\sin\theta\cos\theta + \cos\theta\tan\Phi(\cos\theta - 1)}{\sin\theta(\cos\theta - 1) - \sin^2\theta\tan\Phi} \tag{118}$$

$$e = \frac{K_1(\cos\theta - 1)\tan\Phi + \sin\theta}{\sin\theta(\cos\theta - 1) - \sin^2\theta\tan\Phi} \tag{119}$$

for fixed K_1. We have ellipses and straight lines in (L_0, L_1) space. The major and minor axes of the ellipse are given by b and c respectively. The slope of the line is represented by d and e determines the point at which the line crosses the L_1 axis. The mapping from (K_0, K_2) to (L_0, L_1) and vice versa is given by

$$K_0 = \frac{L_0 - L_1}{2}, \quad K_2 = \frac{L_0 + L_1}{2}. \tag{120}$$

Suppose ω_g is the prescribed closed-loop gain crossover frequency. Then $M_g := \frac{1}{|P(e^{j\theta_g})|}$ and if ϕ_g^* is the desired phase margin in radians, $\Phi_g := \pi + \phi_g^* - \angle P(e^{j\theta_g})$. From (116) we obtain the ellipse and straight lines corresponding to $M = M_g$ and $\Phi = \Phi_g$, giving the design point (K_P^*, K_I^*, K_D^*). If these intersection points lie in the stabilizing set S, the design is feasible, otherwise the specifications have to be altered.

15. Achievable performance with PI and PID controllers

The Gain-Phase Margin design curves represent the achievable performances, that is specified phase and gain margin, that our system can accomplish with a PI or PID controller. The procedure to construct these design curves is the following:

1. Set a test range for phase margins and gain crossover frequencies.
2. For discrete-time PI/PID controllers and continuous-time PI controllers, fix a value of phase margin and gain crossover frequency, plot the corresponding ellipse and straight line following the description in subsection Constant gain and phase loci for a PI Controller. For a continuous-time PID controller, fix a value of phase margin and gain crossover frequency, plot a straight line following the description in subsection the constant gain and phase loci for a PID Controller.
3. If the intersection point of the ellipse and straight line (for discrete-time PI, PID, and continuous-time PI controllers) or the straight line points (for continuous-time PID controller) is contained in the stabilizing set, the point represents the design point with the PI/PID controller gains that satisfy the fixed phase margin and gain crossover frequency. Otherwise, this point is rejected and we go to step 2)

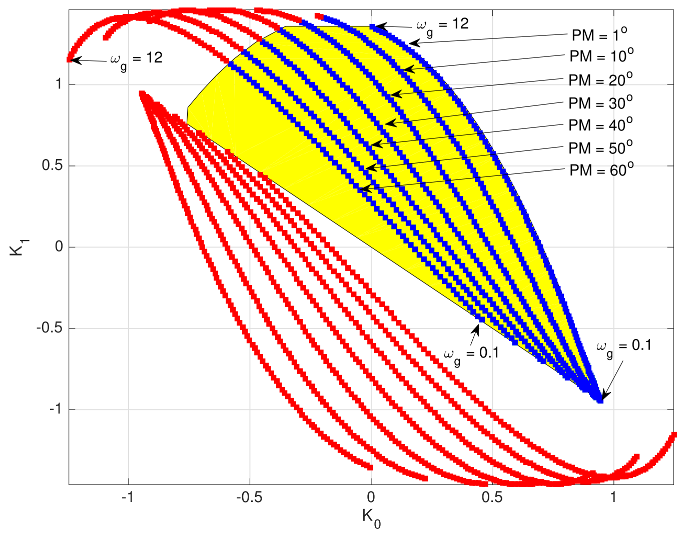


Fig. 27. Stabilizing set in yellow and intersection points of ellipses and straight lines in (K_0, K_1) plane for $PM \in [1, 60]$ degree and $\omega_g \in [0.1, 12]$ rad/s for the discrete-time PI controller Example 15.1. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Given the selected PI/PID controller gains $(K_P^*, K_I^*)/(K_P^*, K_I^*, K_D^*)$, the gain margin of the system is given by

$$GM = \frac{K_P^b}{K_P^*} \tag{121}$$

where K_I^b is the controller gain at the boundary of the stabilizing set following the straight line intersecting the ellipse (for discrete-time PI, PID, and continuous-time PI controllers) or the gain at the boundary of the stabilizing set following the straight line (for continuous-time PID controller).
5. Go to step 2 and repeat for all range of phase margin and gain crossover frequencies.

Example 15.1. Discrete-time PI controller

15.1. Stabilizing set determination

Consider a unity feedback discrete time system with

$$P(z) = \frac{z - 0.1}{z^3 + 0.1z - 0.25} \quad \text{and} \quad C(z) = \frac{K_0 + K_1 z}{z - 1}$$

Following the stabilizing set computation procedure (Bhattacharyya et al., 2009), we obtain the stability region shown in Fig. 27 in the (K_0, K_1) space.

15.2. Gain-phase margin design curves

In Fig. 28 we display an example by selecting a specific phase margin ($PM = 60^o$) for a range of gain crossover frequencies. In Fig. 27, we can see the intersection points of ellipses and straight lines superimposed on the stabilizing set. In this case, the range of gain crossover frequencies is from 0 to 12 rad/s and given achievable phase margins of PM from 1 to 60°.

15.3. Achievable performance

In Fig. 29, we can see the achievable performance for the example. In this case, the maximum gain margin that we can get is 35 dB with a phase margin of 88° with a gain crossover frequency of 0.1 rad/s. The Gain-Phase plane contains information about the capabilities of the system achieving gain margin, phase margins and

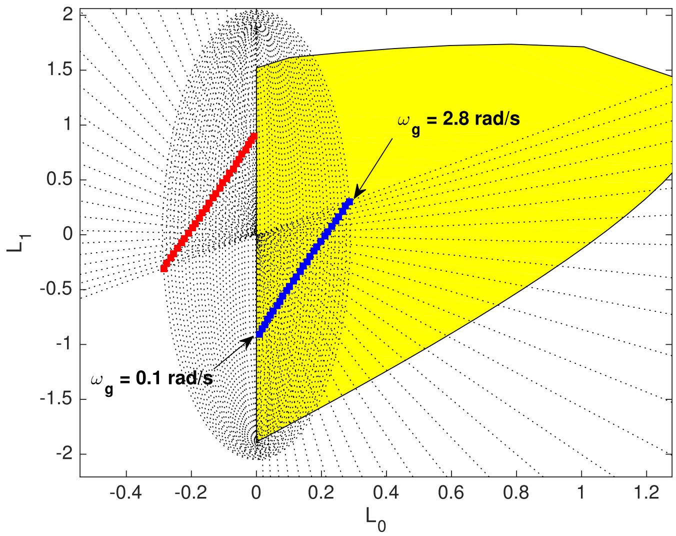


Fig. 28. Stabilizing set in yellow and intersection points of ellipses and straight lines in (L_0, L_1) plane for $PM = 60^o$ and $\omega_g \in [0.1, 2.8]$ rad/s. for the discrete-time PI controller Example 15.1. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

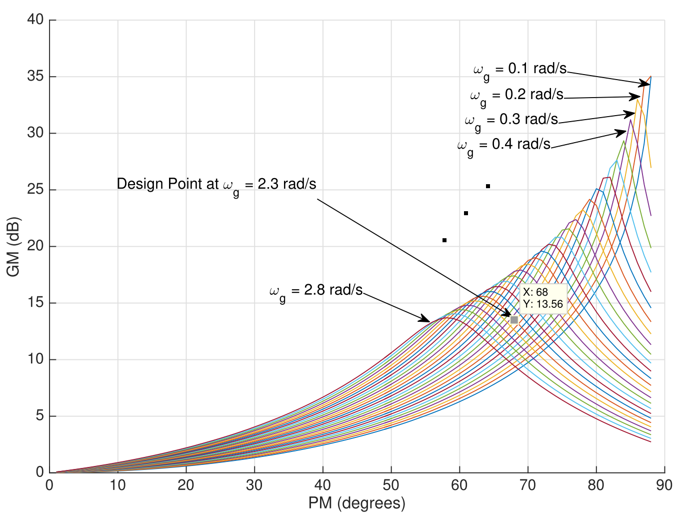


Fig. 29. Achievable gain-phase margin design curves in the Gain-Phase plane for $\omega_g \in [0.1, 2.8]$ rad/s and $PM \in [0, 90]$ degrees for the discrete-time PI controller Example 15.1.

gain crossover frequency simultaneously. Also, it shows the limitations of the system associated with the use of a PI controller.

We can select a design point from Fig. 29. After selecting the point, this design point is found by the intersection of the ellipse and straight line contained in the stabilizing set shown in Fig. 28. In this case, we selected a design point having $PM = 68^o$, $GM = 13.56$ dB, and $\omega_g = 2.3$ rad/s. The PI controller gains for these specifications are

$$K_0^* = -0.06349, \qquad K_1^* = 0.2912 \tag{122}$$

These controller gains correspond to the point of $\omega_g = 2.3$, $PM = 68^o$, and $GM = 13.56$ dB in the Gain-Phase Margin design plane.

Example 15.2. Continuous-Time PI Controller Let us consider a continuous-time LTI unity feedback system represented using the plant

$$P(s) = \frac{s^3 - 4s^2 + s + 2}{s^5 + 8s^4 + 32s^3 + 46s^2 + 46s + 17} \tag{123}$$

and the controller

$$C(s) = \frac{K_P s + K_I}{s} \tag{124}$$

15.4. Computation of the stabilizing set

We follow the procedure summarized in the section to compute the stabilizing set of PI controllers for the given plant. The closed-loop characteristic polynomial is

$$\begin{aligned}\delta(s, K_P, K_I) = {} & s^6 + 8s^5 + (K_P + 32)s^4 + (K_I - 4K_P + 46)s^3 \\ & + (K_P - 4K_I + 46)s^2 + (K_I + 2K_P + 17)s + 2K_I \end{aligned} \tag{125}$$

Here $n = 5$, $m = 3$, and $N(-s) = -s^3 - 4s^2 - s + 2$. Therefore, we obtain

$$\begin{aligned}\nu(s) &= \delta(s, K_P, K_I)N(-s) \\ &= -s^9 - 12s^8 + (-K_P - 65)s^7 + (-K_I - 180)s^6 \\ &\quad + (14K_P - 246)s^5 + (14K_I - 183)s^4 + (-17K_P - 22)s^3 \\ &\quad + (75 - 17K_I)s^2 + (4K_P + 34)s + 4K_I \end{aligned} \tag{126}$$

so that

$$\begin{aligned}\nu(j\omega, K_P, K_I) = {} & -12\omega^8 + (K_I + 180)\omega^6 + (14K_I - 183)\omega^4 \\ & + (17K_I - 75)\omega^2 + 4K_I \\ & + j[-\omega^9 + (K_P + 65)\omega^7 + (14K_P - 246)\omega^5 \\ & + (17K_P + 22)\omega^3 + (4K_P + 34)\omega] \\ = {} & p(\omega) + jq(\omega) \end{aligned} \tag{127}$$

We find that $z^+ = 2$ so that the signature requirement on $\nu(s)$ for stability is,

$$n - m + 1 + 2z^+ = 7 \tag{128}$$

Since the degree of $\nu(s)$ is even, we see from the signature formulas that $q(\omega)$ must have at least three positive real root of odd multiplicity. The range of K_P such that $q(\omega, K_P)$ has at least one real, positive, distinct, finite zero with odd multiplicity was determined to be $K_P \in (-8.5, 4.2)$ which is the allowable range for K_P. By sweeping over different K_P values within the interval $(-8.5, 4.2)$ and following the procedure summarized in the design methodology section, we can generate the set of stabilizing (K_P, K_I) values. This set is shown in Fig. 30.

15.5. Achievable gain-phase margin design curves

As mentioned in Example 15.1, for a range of phase margins and gain crossover frequencies, we superimpose ellipses and straight lines in the stabilizing set. All the intersection points, contained in the stabilizing set, determine the PI controller gains to construct the Gain-Phase design curves shown in Fig. 31. For this example, the evaluated range for phase margin is from 1° to 90° and the range for the gain crossover frequency is from 0.1 to 1 rad/s.

15.6. Achievable performance and final design

As shown in Fig. 31, we notice the achievable performance for Example 15.2. In this case, the maximum gain margin that we can get is 13.13 dB and a phase margin of 79° with a gain crossover frequency of 0.1 rad/s. We can see how when increasing the gain crossover frequency, our values for gain and phase margins decrease. For example, for a given gain crossover frequency of 0.4 rad/s, the maximum gain margin we can get is 2.522 dB and phase margin of 44°. In Fig. 31, it is shown a design point. This design point is found by the intersection of the ellipse and straight line

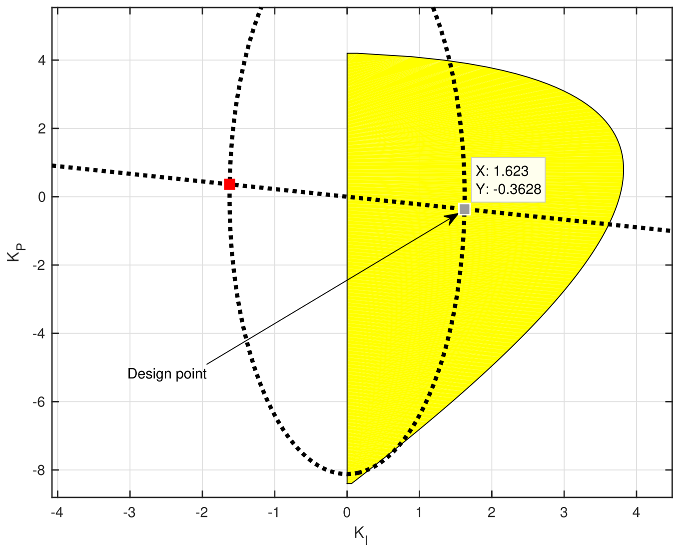


Fig. 30. PI stabilizing set for Example 15.2 and intersection of ellipse and straight line for the final design point.

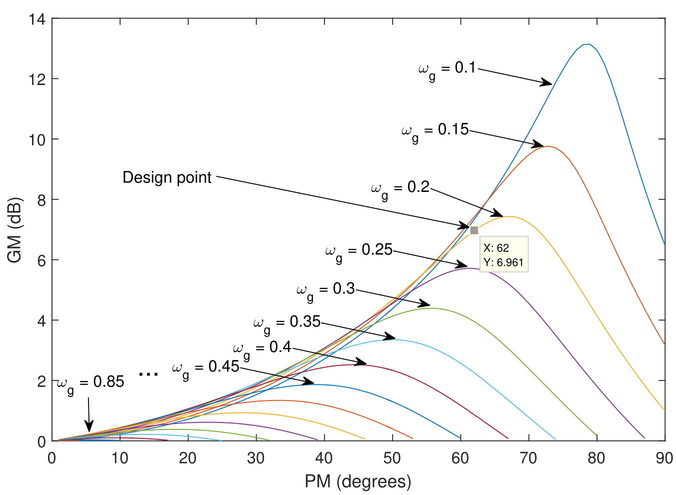


Fig. 31. Achievable Gain-Phase margin design curves in the Gain-Phase plane for Example 15.2.

contained in the stabilizing set shown in Fig. 30. The PI controller gains for these specifications are

$$K_P^* = -0.36283 \tag{129}$$

$$K_I^* = 1.6228 \tag{130}$$

The step response for this controller is given in Fig. 32. These controller gains correspond to the point $\omega_g = 0.2$, $PM = 62^o$, and $GM = 6.96$ dB in the Gain-Phase Margin design plane (see the design point in Fig. 31).

16. Multi-input multi-output (MIMO) control using single-Input single-Output (SISO) methods

In this section we describe a new approach to multivariable control using single-input single-output methods. The details may be found in Mohsenizadeh, Keel, and Bhattacharyya (2015). This has the potential to extend the design capabilities of SISO systems to multivariable systems.

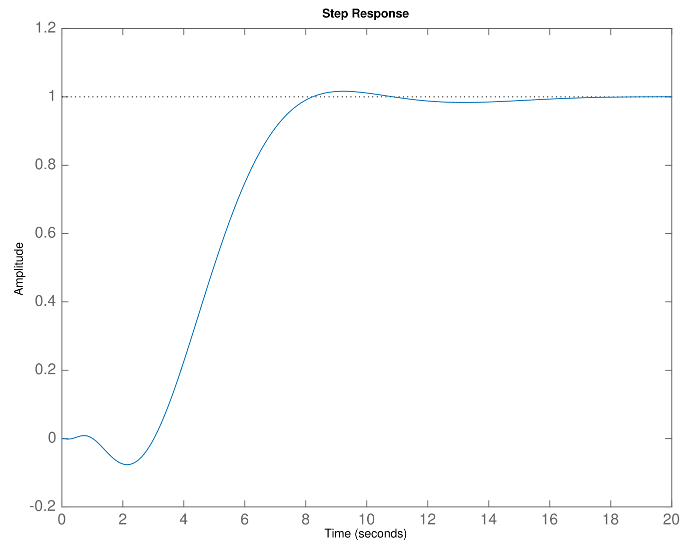


Fig. 32. Step response for the system in Example 15.2 using $C(s)^* = \frac{K_P^* s + K_I^*}{s}$.

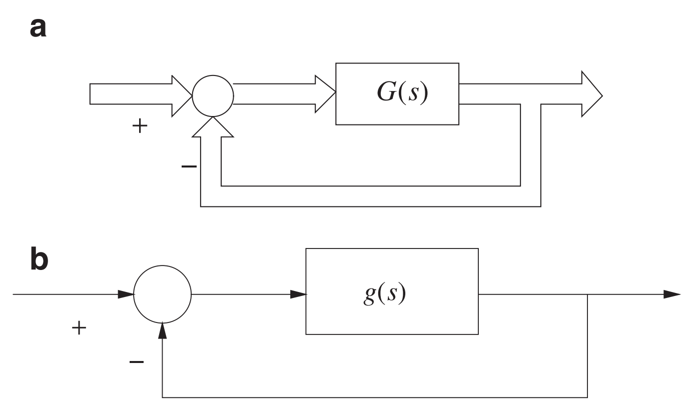


Fig. 33. (a) A multivariable feedback system. (b) A scalar feedback system equivalent to the multivariable system.

16.1. MIMO feedback stability analysis via SISO feedback analysis (Keel & Bhattacharyya, 2015)

16.1.1. Preliminaries

Let $G(s)$ in Fig. 33(a) be an $r \times r$ rational proper transfer function matrix and assume that the feedback system in Fig. 33(a) is well posed. To proceed, introduce the scalar transfer function $g(s)$ constructed from $G(s)$ as follows:

$$g(s) := \sum_{i=1}^{r} m_i(s) \tag{131}$$

where

$$m_i(s) = \text{the sum of the determinants of all } i \times i \text{ leading principal minors of } G(s). \tag{132}$$

Thus, for

$$G(s) = \begin{bmatrix} g_{11}(s) & g_{12}(s) & g_{13}(s) \\ g_{21}(s) & g_{22}(s) & g_{23}(s) \\ g_{31}(s) & g_{32}(s) & g_{33}(s) \end{bmatrix}, \tag{133}$$

we have,

$$g(s) = m_1(s) + m_2(s) + m_3(s) \tag{134}$$

$$m_1(s) = g_{11}(s) + g_{22}(s) + g_{33}(s) \tag{135}$$

$$m_2(s) = \det\begin{bmatrix} g_{11}(s) & g_{12}(s) \\ g_{21}(s) & g_{22}(s) \end{bmatrix} + \det\begin{bmatrix} g_{11}(s) & g_{13}(s) \\ g_{31}(s) & g_{33}(s) \end{bmatrix} + \det\begin{bmatrix} g_{22}(s) & g_{23}(s) \\ g_{32}(s) & g_{33}(s) \end{bmatrix} \tag{136}$$

$$m_3(s) = \det[G(s)]. \tag{137}$$

Now write

$$g(s) = \frac{\nu_g(s)}{\delta_g(s)} \tag{138}$$

where $\nu_g(s)$ and $\delta_g(s)$ are coprime and $\delta_g(s)$ is a monic polynomial. The zeros of $\delta_g(s)$ then are the poles of $g(s)$.

Let also $\delta_G(s)$ denote the monic characteristic polynomial of $G(s)$, that is the monic least common multiple of the denominators of the determinants of all minors of $G(s)$.

16.1.2. Main result

The main result is stated in terms of the scalar unity feedback system in Fig. 33(b) where $g(s)$ is defined as in (131)–(138).

Theorem 16.1. *Mohsenizadeh et al. (2015) The closed-loop characteristic roots of the multivariable system in* Fig. 33(a), consists of the closed-loop characteristic roots of the scalar system in *Fig. 33(b) along with those open-loop characteristic roots of the multivariable system counting multiplicities, that is, roots of $\delta_G(s)$, that are not poles of $g(s)$.*

16.1.3. Stability margin calculations

The result can be easily extended to calculate gain margin and phase margin of the closed-loop system. It follows by the following consideration. Assume that the closed-loop system in Fig. 33(a) (equivalently, Fig. 33(b)) is stable. Let

$$\mathbf{\Delta} := \begin{bmatrix} \Delta & & \\ & \ddots & \\ & & \Delta \end{bmatrix} \tag{139}$$

with an appropriate dimension and consider

$$\det[I + \mathbf{\Delta} G(s)] = 1 + g_\Delta(s) \tag{140}$$

where

$$g_\Delta(s) = \sum_{i=1}^{r} m_i^\Delta(s) \tag{141}$$

and $m_i^\Delta(s)$ is the sum of the determinants of all $i \times i$ leading principal minors of $\mathbf{\Delta} G(s)$. The elements of $\mathbf{\Delta}$ may be replaced by $\Delta = k$ for gain margin, $\Delta = e^{j\theta}$ for phase margin, and $\Delta = e^{-sT}$ for time-delay margin.

Example 16.2. Consider TITO plant $G(s)$ with the transfer function:

$$G(s) = \begin{bmatrix} g_{11}(s) & g_{12}(s) \\ g_{21}(s) & g_{22}(s) \end{bmatrix} = \frac{1}{1.25(s+1)(s+2)}\begin{bmatrix} s-1 & s \\ -6 & s-2 \end{bmatrix}.$$

Then we have

$$g(s) = g_{11}(s) + g_{22}(s) + \det\begin{bmatrix} g_{11}(s) & g_{12}(s) \\ g_{21}(s) & g_{22}(s) \end{bmatrix} = \frac{1.6s - 1.76}{s^2 + 3s + 2}.$$

The Nyquist plot of $g(s)$ is shown in Fig. 34 and the MIMO closed-loop system is stable.

16.1.4. Gain margin calculation

Let $\Delta = k$, then

$$g_\Delta(s) = kg_{11}(s) + kg_{22}(s) + \det\begin{bmatrix} kg_{11}(s) & kg_{12}(s) \\ kg_{21}(s) & kg_{22}(s) \end{bmatrix}.$$

The ranges of the stabilizing gain calculated are:

$$\left\{\begin{array}{c} -1.89 < k < 0 \\ 0 < k < 1.25 \\ 2.5 < k < \infty. \end{array}\right\}$$

Nyquist Plots of $g_\Delta(s)$ for $k = -1.89$, $k = 1.25$, and $k = 2.5$ are given in Fig. 35(a), (b), and (c), respectively.

16.1.5. Phase margin calculation

Let $\Delta = e^{j\theta}$, then

$$g_\Delta(s) = e^{j\theta}g_{11}(s) + e^{j\theta}g_{22}(s) + \det\begin{bmatrix} e^{j\theta}g_{11}(s) & e^{j\theta}g_{12}(s) \\ e^{j\theta}g_{21}(s) & e^{j\theta}g_{22}(s) \end{bmatrix}.$$

The phase margin calculated is:

$0^o < \theta < 40^o$.

The Nyquist Plot of $g_\Delta(s)$ for $\theta = 40$ is given in Fig. 36.

16.1.6. Time-delay margin calculation

Let $\Delta = e^{-sT}$, then

$$g_\Delta(s) = e^{-sT}g_{11}(s) + e^{-sT}g_{22}(s) + \det\begin{bmatrix} e^{-sT}g_{11}(s) & e^{-sT}g_{12}(s) \\ e^{-sT}g_{21}(s) & e^{-sT}g_{22}(s) \end{bmatrix}$$

The time-delay margin calculated is:

$0 < T < 3.2$.

The Nyquist Plot of $g_\Delta(s)$ for $T = 3.2$ is given in Fig. 37(b).

16.2. SISO control of MIMO systems using the Smith–Mcmillan form

Let us denote by $P(s)$ the transfer function matrix of a linear MIMO plant. Suppose that $P(s)$ is an $n \times n$ matrix,

$$P(s) = \begin{bmatrix} p_{11}(s) & \dots & p_{1n}(s) \\ \vdots & \ddots & \vdots \\ p_{n1}(s) & \dots & p_{nn}(s) \end{bmatrix}, \tag{142}$$

where each transfer function $p_{ij}(s)$, $i, j = 1, 2, \dots, n$ is rational and proper. $P(s)$ can be written as

$$P(s) = \frac{1}{d(s)} N(s), \tag{143}$$

where $d(s)$ is the least common multiple of the denominators of all the elements in $P(s)$, and $N(s)$ is a polynomial matrix. Pre-multiplication and post-multiplication of $N(s)$ by appropriate choices of unimodular matrices results in an equivalent diagonal polynomial matrix $S(s)$, known as the Smith form,

$$S(s) = \underbrace{Y_y(s)\dots Y_2(s)Y_1(s)}_{Y(s)} N(s) \underbrace{U_1(s)U_2(s)\dots U_u(s)}_{U(s)}. \tag{144}$$

A unimodular polynomial matrix is a square polynomial matrix whose inverse is also a polynomial matrix. A necessary and sufficient condition for this is that the determinant of the polynomial matrix be a constant. Therefore, $S(s)$ can be written as

$$S = \mathrm{diag}[\epsilon_1'(s), \epsilon_2'(s), \dots, \epsilon_n'(s)], \tag{145}$$

where

$$\epsilon_i'(s) | \epsilon_{i+1}'(s), \quad i = 1, 2, \dots, n-1, \tag{146}$$

meaning that each polynomial $\epsilon_i'(s)$ divides $\epsilon_{i+1}'(s)$ for $i = 1, 2, \dots, n-1$. Dividing each diagonal element of $S(s)$, $\epsilon_i'(s)$, by $d(s)$

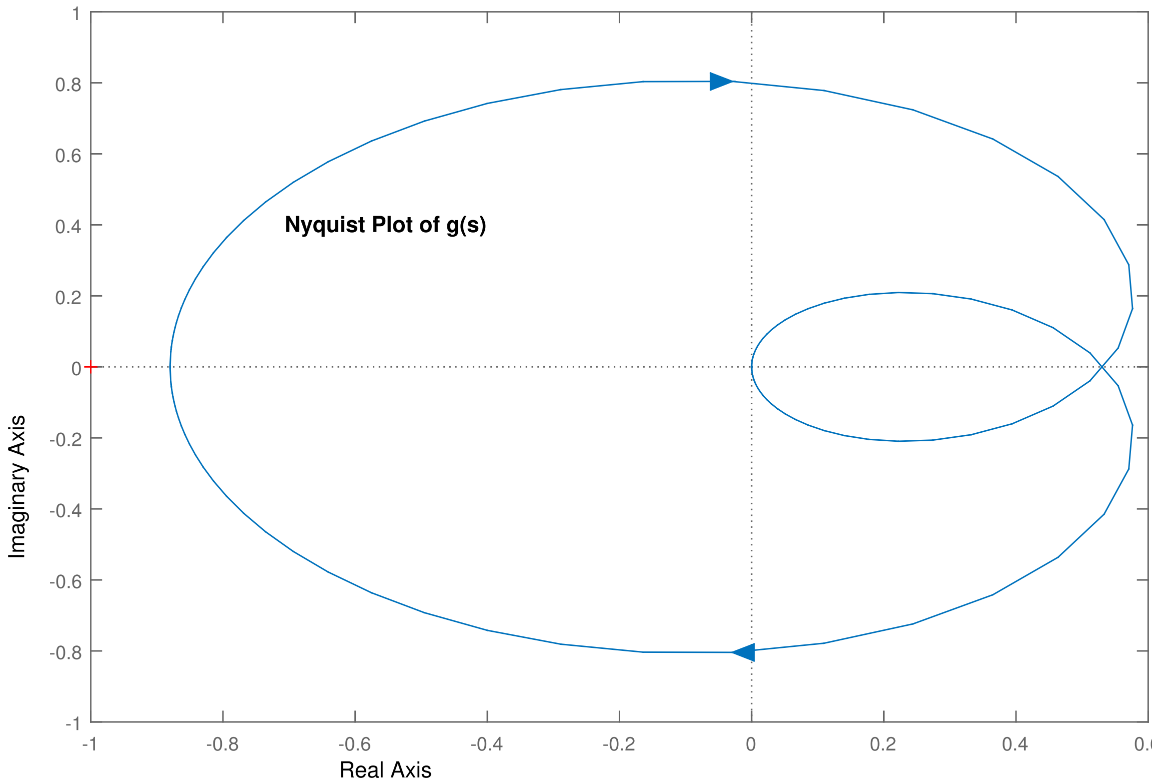


Fig. 34. Nyquist plot of $g(s)$ (Example 16.2).

and performing all possible cancellations yields coprime polynomials $\epsilon_i(s)$ and $\psi_i(s)$ such that

$$\frac{\epsilon_i(s)}{\psi_i(s)} = \frac{\epsilon_i'(s)}{d(s)}, \quad i = 1, 2, \ldots, n, \tag{147}$$

and

$$\begin{aligned}&\epsilon_i(s)|\epsilon_{i+1}(s),\\ &\psi_{i+1}(s)|\psi_i(s), \quad i = 1, 2, \ldots, n-1.\end{aligned} \tag{148}$$

The Smith–McMillan form of the transfer function matrix $P(s)$ will be

$$P_d(s) = \mathrm{diag}\left[\frac{\epsilon_1(s)}{\psi_1(s)}, \frac{\epsilon_2(s)}{\psi_2(s)}, \ldots, \frac{\epsilon_n(s)}{\psi_n(s)}\right]. \tag{149}$$

The poles of $P(s)$ are the roots of the following polynomial, known as the pole polynomial $p(s)$,

$$p(s) := \psi_1(s)\psi_2(s)\cdots\psi_n(s). \tag{150}$$

Similarly, the roots of the zero polynomial $z(s)$,

$$z(s) := \epsilon_1(s)\epsilon_2(s)\cdots\epsilon_n(s), \tag{151}$$

are the zeros of $P(s)$.

16.2.1. Main result

Consider the linear MIMO plant P shown in Fig. 38 where the inputs and outputs are related by

$$y(s) = P(s)u(s). \tag{152}$$

The control problem is to design a multivariable controller $C(s)$ so that the closed loop system, shown in Fig. 39, is stable and the output signals track the reference signals. Let us denote the Smith–McMillan form of the transfer function matrix $P(s)$ by $P_d(s)$. Thus, we can write

$$P_d(s) = Y(s)P(s)U(s), \tag{153}$$

or

$$P(s) = Y^{-1}(s)P_d(s)U^{-1}(s), \tag{154}$$

where $Y(s)$ and $U(s)$ are appropriate unimodular matrices. Note that $Y^{-1}(s)$ and $U^{-1}(s)$ are unimodular polynomial matrices as well. Using (152) and (154) we obtain

$$y(s) = Y^{-1}(s)P_d(s)U^{-1}(s)u(s), \tag{155}$$

and by pre-multiplying (155) by $Y(s)$,

$$\underbrace{Y(s)y(s)}_{y_d(s)} = P_d(s)\underbrace{U^{-1}(s)u(s)}_{u_d(s)}, \tag{156}$$

where we define

$$u_d(s) := U^{-1}(s)u(s), \tag{157}$$

$$y_d(s) := Y(s)y(s). \tag{158}$$

A diagonal controller $C_d(s)$ is now to be designed for the equivalent diagonal system $P_d(s)$ as depicted in Fig. 40. Since the Smith–McMillan form $P_d(s)$ is of diagonal form, one can design a SISO controller for each single loop, considering each diagonal element of $P_d(s)$, independently. The transfer function matrix of the designed controller $C_d(s)$ can be written as

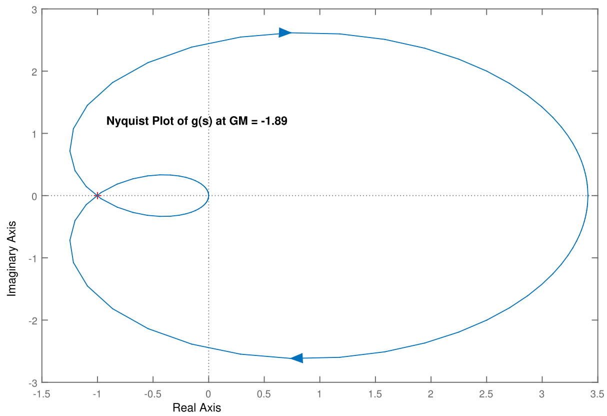
$$C_d(s) = \mathrm{diag}\left[C_1^d(s), C_2^d(s), \ldots, C_n^d(s)\right], \tag{159}$$

where each designed controller $C_k^d(s),\ k = 1, 2, \ldots, n$ has a relative degree r_k^d. Let us define

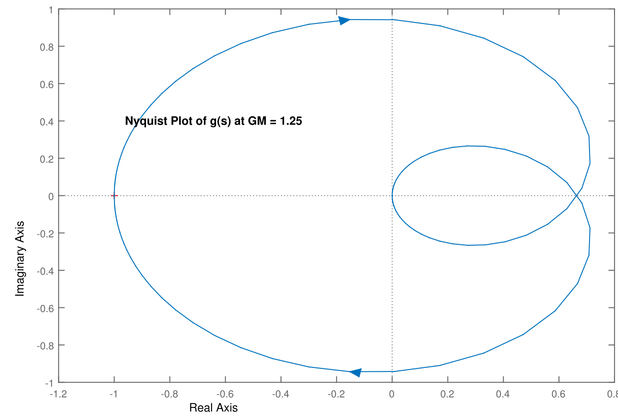
$$r_d(s) := Y(s)r(s). \tag{160}$$

From the closed loop system in Fig. 40 we have

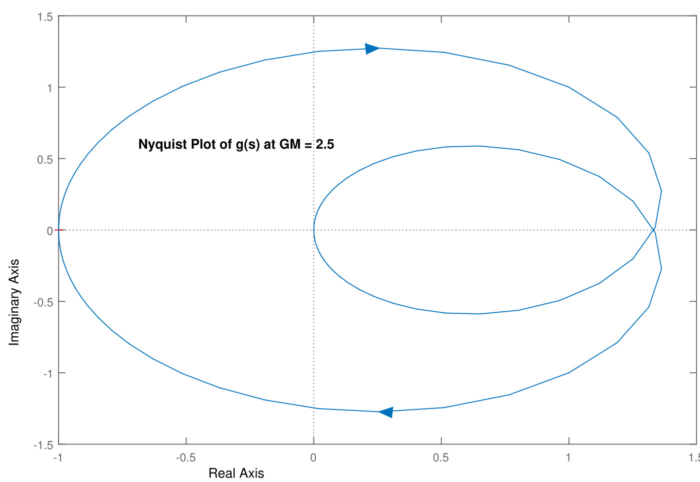
$$e_d(s) = r_d(s) - y_d(s), \tag{161}$$



(a) Nyquist Plot of $g_\Delta(s)$ for $k = -1.89$



(b) Nyquist Plot of $g_\Delta(s)$ for $k = 1.25$



(c) Nyquist Plot of $g_\Delta(s)$ for $k = 2.5$

Fig. 35. Gain margin problem (Example 16.2).

or using (158) and (160),

$$\begin{aligned} e_d(s) &= Y(s)r(s) - Y(s)y(s) = Y(s)(r(s) - y(s)) \\ &= Y(s)e(s). \end{aligned} \tag{162}$$

The inputs and outputs of C_d are related through

$$u_d(s) = C_d(s)e_d(s), \tag{163}$$

which can be written as

$$U^{-1}(s)u(s) = C_d(s)Y(s)e(s), \tag{164}$$

using the definitions in (157) and (162). Pre-multiplying (164) by $U(s)$ yields

$$u(s) = \underbrace{U(s)C_d(s)Y(s)}_{C(s)}e(s), \tag{165}$$

where

$$C(s) := U(s)C_d(s)Y(s) = \begin{bmatrix} C_{11}(s) & \dots & C_{1n}(s) \\ \vdots & \ddots & \vdots \\ C_{n1}(s) & \dots & C_{nn}(s) \end{bmatrix}, \tag{166}$$

is a non-diagonal multivariable controller to be connected to $P(s)$ (see Fig. 39). From (166) it is clear that

$$C_d(s) = U^{-1}(s)C(s)Y^{-1}(s). \tag{167}$$

We introduce the following 4 lemmas leading us toward the design of a diagonal controller C_d that guarantees stability, asymptotic tracking and properness of the corresponding multivariable controller C.

Lemma 16.3. Mohsenizadeh et al. (2015) *The multivariable controller $C(s)$ stabilizes the MIMO plant $P(s)$ if and only if the designed diagonal controller $C_d(s)$ stabilizes the equivalent diagonal plant $P_d(s)$.*

Remark 16.4. For $y_{d_k}(t)$ to track $r_{d_k}(t)$, the controller $C_k^d(s)$ can be designed as

$$C_k^d(s) = \frac{\alpha_k(s)}{\beta_k(s)\gamma_k(s)}, \tag{168}$$

where the characteristic polynomial

$$\psi_k(s)\beta_k(s)\gamma_k(s) + \epsilon_k(s)\alpha_k(s), \tag{169}$$

is Hurwitz, and $\beta_k(s)$ has roots at the poles of $r_{d_k}(s)$, and the product $\beta_k(s)\gamma_k(s)$ has no RHP cancellations with $\epsilon_k(s)$.

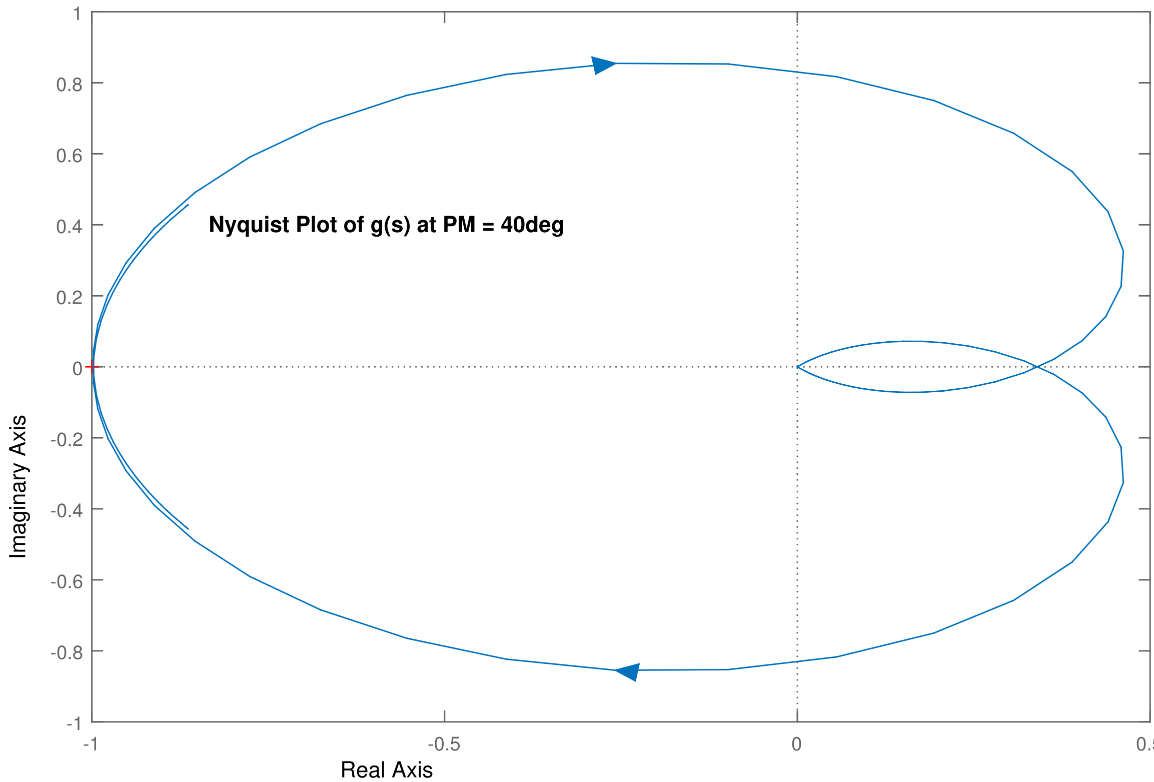


Fig. 36. Nyquist Plot of $g_{\Delta}(s)$ for $\theta = 40^o$ (Phase margin problem: Example 16.2).

Fig. 37. Nyquist Plot of $g_{\Delta}(s)$ for $T = 3.2$ (Time-delay margin problem: Example 16.2).

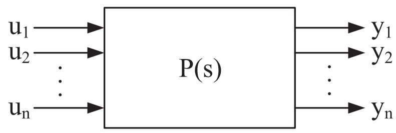


Fig. 38. A linear MIMO plant.

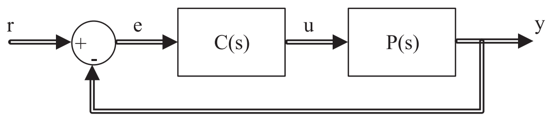


Fig. 39. A MIMO closed loop system with a multivariable controller.

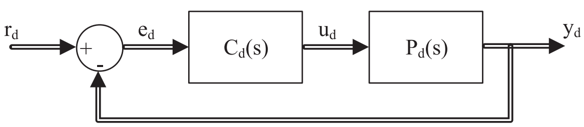


Fig. 40. Closed loop system with the designed diagonal controller.

Let us write the unimodular matrix $Y(s)$, in (153), as

$$Y(s) = \begin{bmatrix} y_{11}(s) & \dots & y_{1n}(s) \\ \vdots & \ddots & \vdots \\ y_{n1}(s) & \cdots & y_{nn}(s) \end{bmatrix}, \tag{170}$$

where $y_{kj}(s),\ k, j = 1, 2, \dots, n$ is a polynomial of degree d^y_{kj}. Similarly, each entry of $U(s)$ in (153) is a polynomial, $u_{ik}(s),\ i, k = 1, 2, \dots, n$ of degree d^u_{ik}.

The following lemma characterizes the required relative degree for each designed SISO controller $C^d_k(s),\ k = 1, 2, \dots, n$, so that the corresponding multivariable controller $C(s)$ is proper.

Lemma 16.5. Mohsenizadeh et al. (2015) *If the relative degrees r^d_k of the designed SISO controllers $C^d_k(s),\ k = 1, 2, \dots, n$, satisfy*

$$\min_{k=1,2,\dots,n} \{r^d_k - d^u_{ik} - d^y_{kj}\} \ge 0, \qquad \forall i, j = 1, 2, \dots, n, \tag{171}$$

where d^u_{ik} and d^y_{kj} are the degree of polynomials in the unimodular matrices $U(s)$ and $Y(s)$, respectively, then the corresponding multivariable controller $C(s)$ will be proper. For any k that $u_{ik}(s)C^d_k(s)y_{kj}(s) = 0$, then the corresponding $r^d_k - d^u_{ik} - d^y_{kj}$ term in (171) should be neglected.

Now, based on Lemmas 16.3–16.5, we can present our main theorem.

Theorem 16.6. Mohsenizadeh et al. (2015) *Given a rational proper transfer function matrix $P(s)$, of a linear MIMO plant, if a diagonal controller $C_d(s)$ is designed such that it stabilizes the equivalent Smith–McMillan form of $P(s)$, denoted by $P_d(s)$, and the output $y_d(t)$ tracks the reference $r_d(t)$, and $C_d(s)$ satisfies the relative degree conditions given in (171), then the corresponding multivariable controller $C(s)$ will be proper and stabilize $P(s)$, and the output $y(t)$ will track the reference $r(t)$.*

Example 16.7. *A two-input two-output (TITO) stable plant*

Consider the following transfer function matrix, see Mohsenizadeh et al. (2015)

$$P(s) = \begin{bmatrix} \frac{4}{(s+1)(s+2)} & \frac{-1}{s+1} \\ \frac{2}{s+1} & \frac{-1}{2(s+1)(s+2)} \end{bmatrix}, \tag{172}$$

representing a TITO stable plant and suppose that a multivariable controller is to be designed so that the closed loop system is stable and the output signals track unit step reference signals.

The least common multiple of the denominators of all the elements in $P(s)$ is

$$d(s) = (s+1)(s+2). \tag{173}$$

Thus, we can write $P(s)$ as

$$P(s) = \frac{1}{(s+1)(s+2)} \underbrace{\begin{bmatrix} 4 & -s-2 \\ 2s+4 & \frac{-1}{2} \end{bmatrix}}_{N(s)}. \tag{174}$$

The Smith form of $N(s)$ can be obtained by multiplying $N(s)$ by the following unimodular matrices $Y(s)$ and $U(s)$,

$$\begin{aligned} S(s) &= \underbrace{\begin{bmatrix} \frac{1}{4} & 0 \\ -s-2 & 2 \end{bmatrix}}_{Y(s)} \underbrace{\begin{bmatrix} 4 & -s-2 \\ 2s+4 & \frac{-1}{2} \end{bmatrix}}_{N(s)} \underbrace{\begin{bmatrix} 1 & \frac{s+2}{4} \\ 0 & 1 \end{bmatrix}}_{U(s)} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & s^2+4s+3 \end{bmatrix}. \end{aligned} \tag{175}$$

For the above unimodular matrices, we have $d^u_{11} = d^u_{21} = d^u_{22} = d^y_{11} = d^y_{12} = d^y_{22} = 0$ and $d^u_{12} = d^y_{21} = 1$. Dividing each element of $S(s)$, in (175), by $d(s)$, in (173), and performing all possible cancellations gives the Smith–McMillan form of $P(s)$ as

$$P_d(s) = \begin{bmatrix} \frac{1}{(s+1)(s+2)} & 0 \\ 0 & \frac{s+3}{s+2} \end{bmatrix}. \tag{176}$$

Now, two SISO controllers $C^d_1(s)$ and $C^d_2(s)$ can be designed for two single loops corresponding to the diagonal elements of $P_d(s)$, to satisfy closed loop stability and reference signal tracking. Let us consider the following controller

$$C_d(s) = \begin{bmatrix} \frac{3}{s(s+4)} & 0 \\ 0 & \frac{1}{2s(s+1)} \end{bmatrix}, \tag{177}$$

where both relative degrees r^d_1 and r^d_2 are 2. Based on the statement of Lemma 16.5, if r^d_1 and r^d_2 satisfy the following set of inequalities,

$$\begin{aligned} &\min_{k=1,2}\{r^d_k - d^u_{1k} - d^y_{k1}\} \ge 0, \\ &\min_{k=1,2}\{r^d_k - d^u_{1k} - d^y_{k2}\} \ge 0, \\ &\min_{k=1,2}\{r^d_k - d^u_{2k} - d^y_{k1}\} \ge 0, \\ &\min_{k=1,2}\{r^d_k - d^u_{2k} - d^y_{k2}\} \ge 0, \end{aligned} \tag{178}$$

then the multivariable controller $C(s)$ will be proper. Expanding the terms in (178) and substituting the values for d^u, d^y and r^d, it can be easily verified that the above set of inequalities are satisfied. The transfer function matrix $C(s)$ is

$$C(s) = U(s)C_d(s)Y(s) = \begin{bmatrix} \dfrac{-(s^3+8s^2+14s+10)}{8s(s+1)(s+4)} & \frac{s+2}{4s(s+1)} \\ \dfrac{-(s+2)}{2s(s+1)} & \frac{1}{s(s+1)} \end{bmatrix}, \tag{179}$$

which is proper. Connecting $C(s)$ to $P(s)$ and closing the loop, the closed loop system transfer function matrix $H(s)$ becomes

$$\begin{aligned} H(s) &= [I + P(s)C(s)]^{-1}P(s)C(s) \\ &= \begin{bmatrix} \frac{3}{s^4+7s^3+14s^2+8s+3} & 0 \\ \frac{-s(s+2)(s^4+10s^3+29s^2+32s+12)}{2(s^4+7s^3+14s^2+8s+3)(2s^3+6s^2+5s+3)} & \frac{s+3}{2s^3+6s^2+5s+3} \end{bmatrix}. \end{aligned} \tag{180}$$

The response of the closed loop system to unit steps is plotted in Fig. 41 and verifies tracking of independent step inputs.

Example 16.8. *A two-input two-output (TITO) unstable plant*

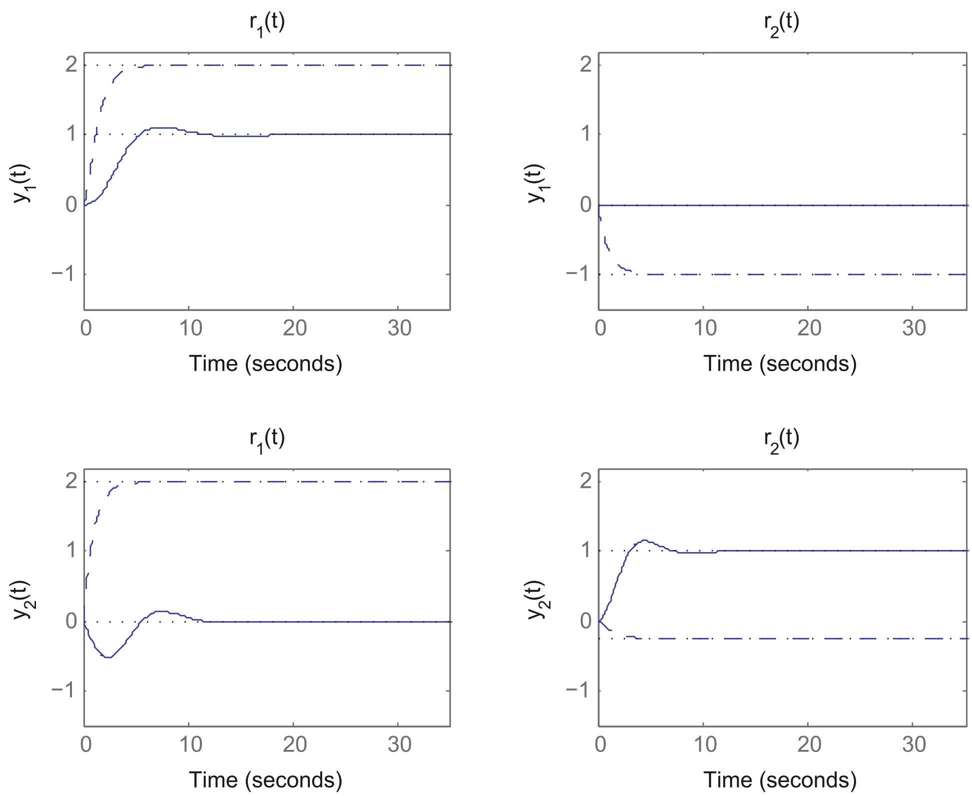


Fig. 41. Step response of the closed loop system after connecting $C(s)$ in (179) to $P(s)$ in (172) (solid line), and the step response of the open loop plant $P(s)$ in (172) (dashed line).

Consider the following TITO unstable transfer function matrix

$$P(s) = \begin{bmatrix} \frac{4}{s-1} & \frac{-1}{s+1} \\ \frac{2}{s+1} & \frac{1}{(s-1)} \end{bmatrix}. \tag{181}$$

Our objective is to design a multivariable controller $C(s)$ to stabilize the closed loop system and also make the outputs track unit steps. Here, we have $d(s) = (s+1)(s-1)$. Thus, $P(s)$ can be written as

$$P(s) = \frac{1}{(s+1)(s-1)} \underbrace{\begin{bmatrix} 4s+4 & -s+1 \\ 2s-2 & s+1 \end{bmatrix}}_{N(s)}, \tag{182}$$

and the Smith form of $N(s)$ can be calculated as follows

$$S(s) = \underbrace{\begin{bmatrix} \frac{1}{8} & \frac{-1}{4} \\ \frac{-s+1}{3} & \frac{2s+2}{3} \end{bmatrix}}_{Y(s)} \underbrace{\begin{bmatrix} 4s+4 & -s+1 \\ 2s-2 & s+1 \end{bmatrix}}_{N(s)} \underbrace{\begin{bmatrix} 1 & \frac{3s+1}{8} \\ 0 & 1 \end{bmatrix}}_{U(s)}$$
$$= \begin{bmatrix} 1 & 0 \\ 0 & s^2 + \frac{2}{3}s + 1 \end{bmatrix}. \tag{183}$$

For this example, we have $d^u_{11} = d^u_{21} = d^u_{22} = d^y_{11} = d^y_{12} = 0$ and $d^u_{12} = d^y_{21} = d^y_{22} = 1$. The Smith–McMillan form of $P(s)$ will be

$$P_d(s) = \begin{bmatrix} \frac{1}{(s^2-1)} & 0 \\ 0 & \frac{3s^2+2s+3}{3(s^2-1)} \end{bmatrix}. \tag{184}$$

Now, two SISO stabilizing controllers $C^d_1(s)$ and $C^d_2(s)$ should be designed for the diagonal elements of $P_d(s)$ in (184) that also guarantee unit step tracking. Moreover, the relative degrees of these controllers need to satisfy the inequality conditions given in the statement of Lemma 16.5. Consider the following controller

$$C_d(s) = \begin{bmatrix} \frac{5s^2+5s+1}{s(.1s+1)} & 0 \\ 0 & \frac{5}{s(.1s+1)} \end{bmatrix}, \tag{185}$$

where the relative degrees of $C^d_1(s)$ and $C^d_2(s)$ meet the inequality conditions in Lemma 16.5. The multivariable controller $C(s)$ becomes

$$C(s) = \begin{bmatrix} \frac{1.05s+.336}{s(.1s+1)} & \frac{.415s+.166}{s(.1s+1)} \\ \frac{-1.67(s-1)}{s(.1s+1)} & \frac{3.33(s+1)}{s(.1s+1)} \end{bmatrix}, \tag{186}$$

which is strictly proper, and the closed loop system transfer function matrix $H(s)$ will be

$$H(s) = \begin{bmatrix} \frac{h_{11}(s)}{g(s)} & \frac{h_{12}(s)}{g(s)} \\ \frac{h_{21}(s)}{g(s)} & \frac{h_{22}(s)}{g(s)} \end{bmatrix}, \tag{187}$$

where

$$\begin{aligned} g(s) &= s^8 + 20s^7 + 198s^6 + 1043s^5 + 3094s^4 \\ &\quad + 3703s^3 + 3873s^2 + 2233s + 500, \\ h_{11}(s) &= 58s^6 + 605s^5 + 2688s^4 + 3862s^3 \\ &\quad + 4420s^2 + 1533s + 500, \\ h_{12}(s) &= -s(17s^5 + 143s^4 - 290s^3 \\ &\quad - 543s^2 + 273s + 400), \\ h_{21}(s) &= s(4s^5 + 28s^4 - 136s^3 \\ &\quad + 73s^2 + 132s - 100), \\ h_{22}(s) &= 42s^6 + 478s^5 + 3105s^4 + 3988s^3 \\ &\quad + 4020s^2 + 2533s + 500. \end{aligned} \tag{188}$$

The response of this closed loop system to unit steps is plotted in Fig. 42. Since $H(s)|_{s=0} = I$, the closed loop system tracks steps with zero steady state error.

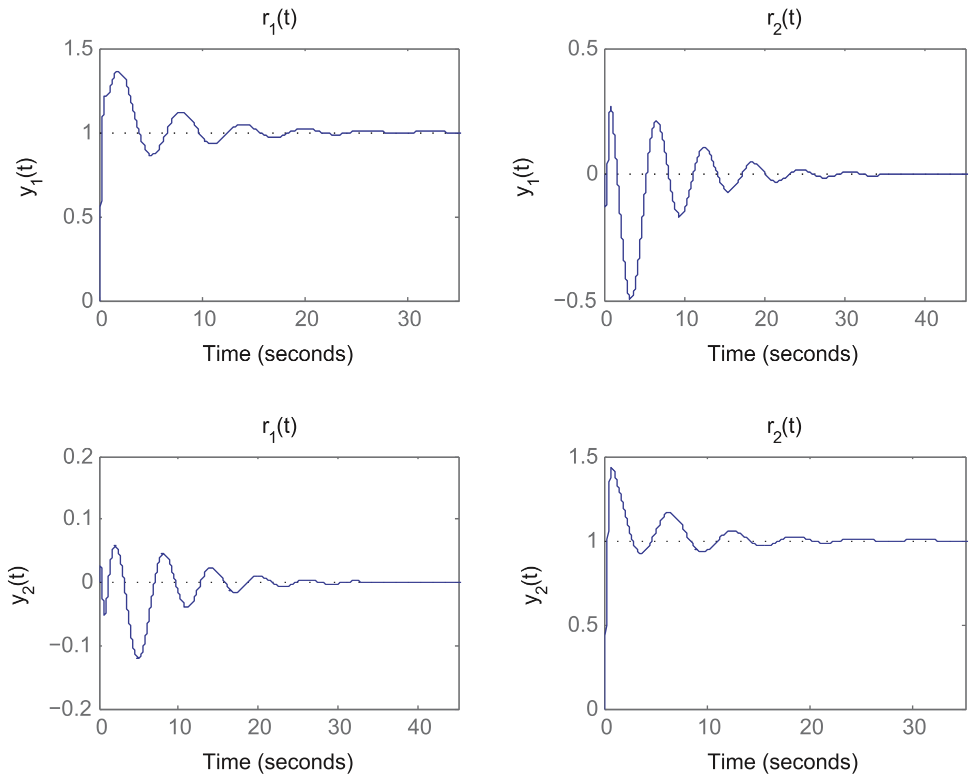


Fig. 42. Step response of the closed loop system after connecting $C(s)$ in (186) to $P(s)$ in (181).

17. Concluding remarks

This relatively brief overview of the subject of robust control under parametric uncertainty is necessarily incomplete and the author apologizes in advance for omissions of content or authorship and any personal bias in choice of topics. It is hoped that this article may be helpful to the reader who wishes to go deeper into specific topics. We have avoided some areas altogether such as W. M. Wonham's geometric theory (Wonham, 1974) and robust adaptive control.

We have compiled an extensive list of references on robustness under parametric uncertainty. In order to save space this is made available to the reader via http://ece.tamu.edu/~bhatt/spbreference2017.html. This database indicates separately the analysis and synthesis results and theory based on Kharitonov's Theorem.

The reader may appreciate, after reading this article, that robustness and its attainment has been a fundamental driving force in control theory, and this will continue to hold as we move into the area of driver-less cars, multiagent systems and advanced robotics. It will be interesting to see how simple controllers, such as PID's, might be adapted and redesigned to tackle the complex tasks to be handled in these environments.

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